

Chapter 42

UART Controller (UART)

42.1 Overview

A UART is a character-oriented data link for asynchronous communication between devices. Such communication does not add clock signals to the data sent. Therefore, in order to communicate successfully, the transmitter and the receiver must operate at the same baud rate with the same stop bit(s) and parity bit(s).

A UART data frame usually begins with one start bit, followed by data bits, one parity bit (optional), and one or more stop bits. UART controllers on ESP32-P4 support various lengths of data bits and stop bits. These controllers also support software and hardware flow control as well as GDMA for high-speed data transfer. This allows developers to use multiple UART ports at minimal software cost.

ESP32-P4 has six UART controllers, including five regular UARTs and one low-power (LP) UART. These UARTs are compatible with various UART devices, and support Infrared Data Association (IrDA) and RS485 communication. In this chapter, the five regular UART controllers are referred to as UART n , in which n denotes 0, 1, 2, 3, or 4. LP UART is the cut-down version of the regular UART, with a separate group of registers. For differences between UART and LP UART, please refer to Table 42.2-1.

42.2 Features

Table 42.2-1 lists the feature comparison between UART and LP UART:

Table 42.2-1. UART and LP UART Feature Comparison

UART Feature	LP UART Feature
Programmable baud rate up to 5 MBaud	
128 x 8 bit RAM respectively for the TX channel and RX channel of a UART controller	20 x 8-bit RAM, shared by the TX FIFO and RX FIFO of LP UART
Full-duplex asynchronous communication	
Data bits (5 to 8 bits)	
Stop bits (1, 1.5, or 2 bits)	
Parity bit	
Special character AT_CMD detection	
RS485 protocol	—
IrDA protocol	—

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UART Feature	LP_UART Feature
High-speed data communication using GDMA	—
Receive timeout	
UART as wakeup source	
Software and hardware flow control	
Three prescalable clock sources: 1. XTAL_CLK 2. RC_FAST_CLK 3. PLL_F80M_CLK	Three prescalable clock sources: 1. RC_FAST_CLK 2. XTAL_DIV_CLK 3. PLL_F8M_CLK

The following description mainly covers regular UART controllers.

42.3 UART Structure

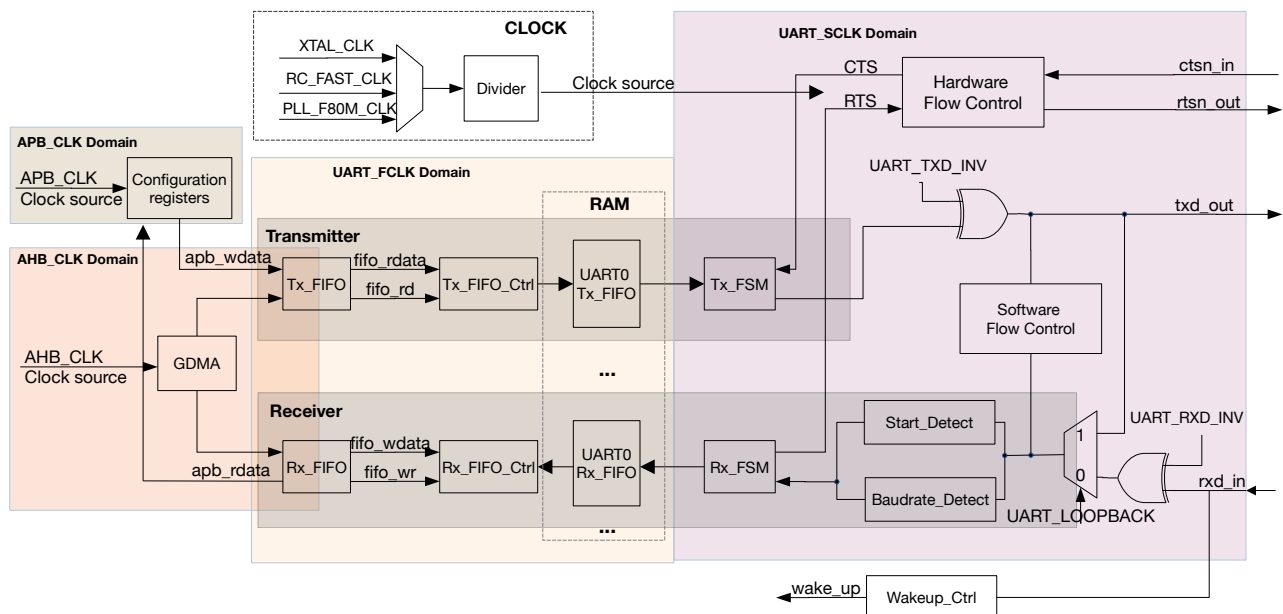


Figure 42.3-1. UART Structure

Figure 42.3-1 shows the basic structure of a UART controller. A UART controller works in four clock domains, namely APB_CLK, AHB_CLK, UART_SCLK, and UART_FCLK. APB_CLK and AHB_CLK are synchronized but with different frequencies (APB_CLK is derived from AHB_CLK by division), and likewise UART_SCLK and UART_FCLK are synchronized but with different frequencies (UART_SCLK is derived from UART_FCLK by division). UART_FCLK has three clock sources: PLL_F80M_CLK, RC_FAST_CLK, and crystal clock XTAL_CLK (for details, please refer to Chapter 10 *Reset and Clock*), which are selected by configuring `HP_SYS_UARTn_CLK_SRC_SEL`. The selected clock source is divided by a divider to generate UART_SCLK clock signals. The divisor is configured by `HP_SYS_CLKRST_UARTn_SCLK_DIV_NUM` for the integral part, `HP_SYS_CLKRST_UARTn_SCLK_DIV_DENOMINATOR` for the denominator of the fractional part, and `HP_SYS_CLKRST_UARTn_SCLK_DIV_NUMERATOR` for the numerator of the fractional part. The divisor ranges from 1 ~ 256. Only regular UART has such a divider; LP UART does not.

A UART controller can be broken down into two parts based on functions: a transmitter and a receiver.

The transmitter contains a TX FIFO (i.e., Tx_FIFO in Figure 42.3-1), which buffers data to be sent. Software can write data to Tx_FIFO via the APB bus, or move data to Tx_FIFO using GDMA. Tx_FIFO_Ctrl controls writing and reading Tx_FIFO. When Tx_FIFO is not empty, Tx_FSM reads data bits in the data frame via Tx_FIFO_Ctrl, and converts them into a bitstream. The levels of output bitstream signal txd_out can be inverted by configuring the [UART_TXD_INV](#) field.

The receiver contains an RX FIFO (i.e., Rx_FIFO in Figure 42.3-1), which buffers data to be processed. The input bitstream signal rxd_in is transferred to the UART controller, and its level can be inverted by configuring [UART_RXD_INV](#) field. Baudrate_Detect measures the baud rate of input bitstream signal rxd_in by detecting its minimum pulse width. Start_Detect detects the start bit in a data frame. If the start bit is detected, Rx_FSM stores data bits in the data frame into Rx_FIFO by Rx_FIFO_Ctrl. Software can read data from Rx_FIFO via the APB bus, or receive data using GDMA.

HW_Flow_Ctrl controls rxd_in and txd_out data flows by standard UART RTS and CTS flow control signals (rtsn_out and ctsn_in). SW_Flow_Ctrl controls data flows by adding special characters to outgoing data and detecting special characters in incoming data. When a UART controller is in Light-sleep mode (see Chapter 14 [Low-Power Management](#) for more details), a wake_up signal can be generated in four ways and sent to RTC, which then wakes up the ESP32-P4 chip. For more information about wakeup, please refer to Section 42.4.8.

42.4 Functional Description

42.4.1 Clock and Reset

UART controllers are asynchronous. Their register configuration module works in the APB_CLK domain. TX FIFO and RX FIFO work across the AHB_CLK and UART_FCLK domains. The UART RAM control unit works in the UART_FCLK domain. The UART transmission and reception control module works in the UART_SCLK domain, i.e., UART Core's clock domain.

When the frequency of the UART_SCLK is higher than the frequency needed to generate the baud rate, the UART Core can be clocked at a lower frequency by the divider, in order to reduce power consumption. Usually, the UART Core's clock frequency is lower than the APB_CLK's frequency, and can be divided by the largest divisor when higher than the frequency needed to generate the baud rate. The frequency of the UART Core's clock can also be at most twice higher than the APB_CLK. The clock for the UART transmitter and the UART receiver can be controlled independently. To enable the clock for the UART transmitter, [UART_TX_SCLK_EN](#) shall be set; to enable the clock for the UART receiver, [UART_RX_SCLK_EN](#) shall be set.

To ensure that the configured register values are synchronized from APB_CLK domain to the UART_SCLK domain, please follow the procedures in Section 42.5.

To reset the whole UART, please:

- Enable the UART Core's clock by setting [HP_SYS_CLKRST_UART_n_CLK_EN](#) to 1.
- Write 1 to [HP_SYS_CLKRST_RST_EN_UART_n_CORE](#).
- Clear [HP_SYS_CLKRST_RST_EN_UART_n_CORE](#) to 0.

42.4.2 UART FIFO

The transmitter and the receiver on the UART controller each use a 128 x 8-bit RAM, and access their respective RAM through a separate 4 x 8-bit asynchronous FIFO interface. The RAM and asynchronous FIFO interface for the transmitter and the receiver are independent and cannot be shared.

UART Tx_FIFO is reset by setting [UART_TXFIFO_RST](#). UART Rx_FIFO is reset by setting [UART_RXFIFO_RST](#).

Data to be sent is written to TX FIFO via the APB bus or using GDMA, read automatically, and converted from a frame into a bitstream by hardware Tx_FSM. Data received is converted from a bitstream into a frame by hardware Rx_FSM, written into RX FIFO, and then stored into RAM via the APB bus or using GDMA. The five UART controllers share one GDMA channel.

The empty signal threshold for Tx_FIFO is configured by setting [UART_TXFIFO_EMPTY_THRHD](#). When data stored in Tx_FIFO is less than [UART_TXFIFO_EMPTY_THRHD](#), a UART_TXFIFO_EMPTY_INT interrupt is generated. The full signal threshold for Rx_FIFO is configured by setting [UART_RXFIFO_FULL_THRHD](#). When data stored in Rx_FIFO is greater than or equal to [UART_RXFIFO_FULL_THRHD](#), a UART_RXFIFO_FULL_INT interrupt is generated. In addition, when Rx_FIFO receives more data than its capacity, a UART_RXFIFO_OVF_INT interrupt is generated.

UART_n can access FIFO via register [UART_FIFO_REG](#). You can put data into TX FIFO by writing [UART_RXFIFO_RD_BYTE](#), and get data in RX FIFO by reading [UART_RXFIFO_RD_BYTE](#).

42.4.3 Baud Rate Generation and Detection

42.4.3.1 Baud Rate Generation

Before a UART controller sends or receives data, the baud rate should be configured by setting corresponding registers. The baud rate generator of a UART controller functions by dividing the input clock source. It can divide the clock source by a fractional amount. The divisor is configured by [UART_CLKDIV_SYNC_REG](#): [UART_CLKDIV](#) for the integral part, and [UART_CLKDIV_FRAG](#) for the fractional part. When using the 80 MHz input clock, the UART controller supports a maximum baud rate of 5 MBaud.

The divisor of the baud rate divider is equal to

$$UART_CLKDIV + \frac{UART_CLKDIV_FRAG}{16}$$

meaning that the final baud rate is equal to

$$\frac{INPUT_FREQ}{UART_CLKDIV + \frac{UART_CLKDIV_FRAG}{16}}$$

where INPUT_FREQ is the frequency of UART Core's source clock. For example, if [UART_CLKDIV](#) = 694 and [UART_CLKDIV_FRAG](#) = 7, then the divisor value is

$$694 + \frac{7}{16} = 694.4375$$

When [UART_CLKDIV_FRAG](#) is 0, the baud rate generator is an integer clock divider where an output pulse is generated every [UART_CLKDIV](#) input pulses.

When [UART_CLKDIV_FRAG](#) is not 0, the divider is fractional and the output baud rate clock pulses are not strictly uniform. As shown in Figure 42.4-1, for every 16 output pulses, the generator divides either

($\text{UART_CLKDIV} + 1$) input pulses or UART_CLKDIV input pulses per output pulse. A total of UART_CLKDIV_FRAG output pulses are generated by dividing ($\text{UART_CLKDIV} + 1$) input pulses, and the remaining ($16 - \text{UART_CLKDIV_FRAG}$) output pulses are generated by dividing UART_CLKDIV input pulses.

The output pulses are interleaved as shown in Figure 42.4-1 below, to make the output timing more uniform:

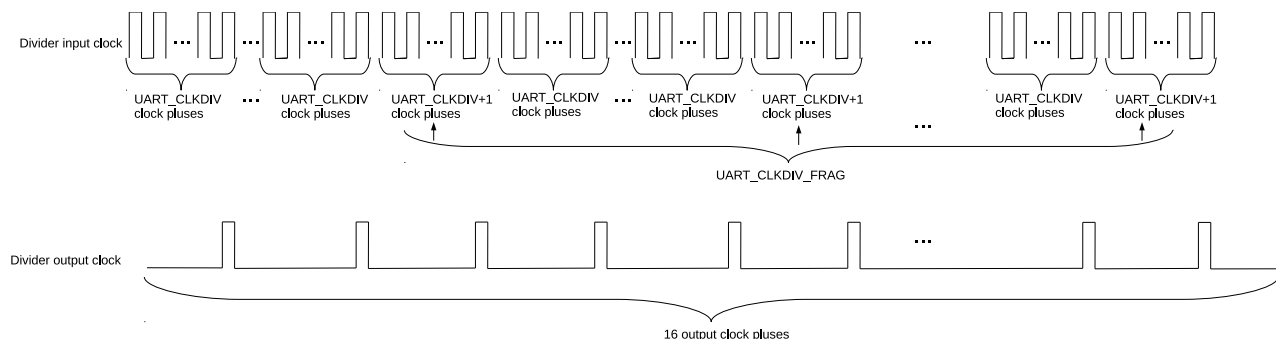


Figure 42.4-1. UART Controllers Division

To support IrDA (see Section 42.4.7 for details), the fractional clock divider for IrDA data transmission generates clock signals divided by $16 \times \text{UART_CLKDIV_SYNC_REG}$. This divider works similarly as the one elaborated above: it takes $\text{UART_CLKDIV}/16$ as the integer value and the lowest four bits of UART_CLKDIV as the fractional value.

42.4.3.2 Baud Rate Detection

Automatic baud rate detection (Autobaud) on UARTs is enabled by setting `UART_AUTOBAUD_EN`. The Baudrate_Detect module shown in Figure 42.4-2 filters any noise whose pulse width is shorter than `UART_GLITCH_FILT`.

Before communication starts, the transmitter could send random data to the receiver for baud rate detection. `UART_LOWPULSE_MIN_CNT` stores the minimum low pulse width, `UART_HIGHPULSE_MIN_CNT` stores the minimum high pulse width, `UART_POSEDGE_MIN_CNT` stores the minimum pulse width between two positive edges, and `UART_NEGEDGE_MIN_CNT` stores the minimum pulse width between two negative edges. These four fields are read by software to determine the transmitter's baud rate.

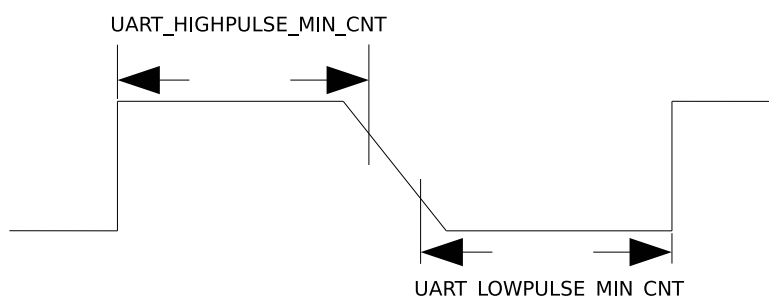


Figure 42.4-2. The Timing Diagram of Weak UART Signals Along Negative Edges

The baud rate can be determined in the following three ways:

1. Normally, to avoid sampling erroneous data along positive or negative edges in a metastable state, which results in the inaccuracy of `UART_LOWPULSE_MIN_CNT` or `UART_HIGHPULSE_MIN_CNT`, use a

weighted average of these two values to eliminate errors for 1-bit pulses. In this case, the baud rate is calculated as follows:

$$B_{\text{uart}} = \frac{f_{\text{clk}}}{(\text{UART_LOWPULSE_MIN_CNT} + \text{UART_HIGHPULSE_MIN_CNT} + 2)/2}$$

- If UART signals are weak along negative edges as shown in Figure 42.4-2, which leads to an inaccurate average of `UART_LOWPULSE_MIN_CNT` and `UART_HIGHPULSE_MIN_CNT`, use `UART_POSEDGE_MIN_CNT` to determine the transmitter's baud rate as follows:

$$B_{\text{uart}} = \frac{f_{\text{clk}}}{(\text{UART_POSEDGE_MIN_CNT} + 1)/2}$$

- If UART signals are weak along positive edges, use `UART_NEGEDGE_MIN_CNT` to determine the transmitter's baud rate as follows:

$$B_{\text{uart}} = \frac{f_{\text{clk}}}{(\text{UART_NEGEDGE_MIN_CNT} + 1)/2}$$

42.4.4 UART Data Frame

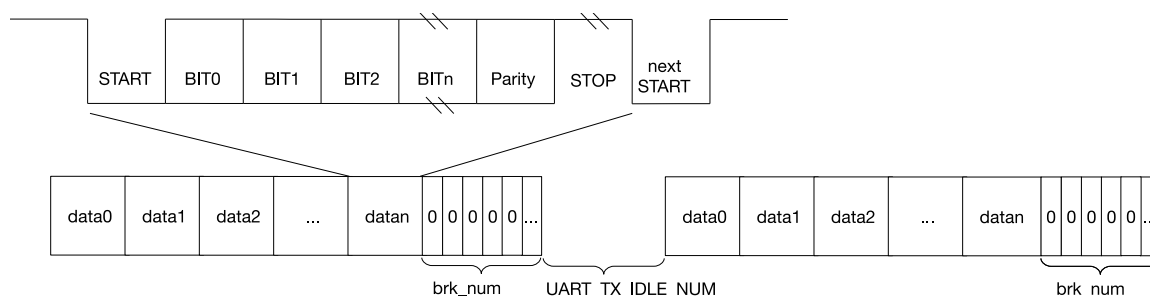


Figure 42.4-3. Structure of UART Data Frame

Figure 42.4-3 shows the basic structure of a data frame. A frame starts with one start bit, and ends with stop bits which can be 1, 1.5, or 2 bit-long, configured by `UART_STOP_BIT_NUM` (in RS485 mode turnaround delay may be added. See details in Section 42.4.6.2). The start bit is logical low, whereas stop bits are logical high.

The actual data length can be anywhere between 5 ~ 8 bit, configured by `UART_BIT_NUM`. When `UART_PARITY_EN` is set, a parity bit is added after data bits. `UART_PARITY` is used to choose even parity or odd parity. When the receiver detects a parity bit error in the data received, a `UART_PARITY_ERR_INT` interrupt is generated, and the data received will still be stored into RX FIFO. When the receiver detects a data frame error, a `UART_FRM_ERR_INT` interrupt is generated, and the data received by default is stored into RX FIFO.

If all data in `Tx_FIFO` has been sent, a `UART_TX_DONE_INT` interrupt is generated. After this, if the `UART_TXD_BRK` bit is set, then the transmitter will enter the break condition and send several NULL characters in which the TX data line is logical low. The number of NULL characters is configured by `UART_TX_BRK_NUM`. Once the transmitter has sent all NULL characters, a `UART_TX_BRK_DONE_INT` interrupt is generated. The minimum interval between data frames can be configured using `UART_TX_IDLE_NUM`. If the transmitter stays idle for `UART_TX_IDLE_NUM` or more time, a `UART_TX_BRK_IDLE_DONE_INT` interrupt is generated.

The receiver can also detect the Break conditions when the RX data line detects any low logical level for one NULL character transmission, and a `UART_BRK_DET_INT` interrupt will be triggered to detect when a break condition has been completed.

The receiver can detect the current bus state through the timeout interrupt `UART_RXFIFO_TOUT_INT`. The `UART_RXFIFO_TOUT_INT` interrupt will be triggered when the bus is in the idle state for more than `UART_RX_TOUT_THRHD` bit time on current baud rate after the receiver has received at least one byte. You can use this interrupt to detect whether all the data from the transmitter has been sent.

42.4.5 AT_CMD Character Structure

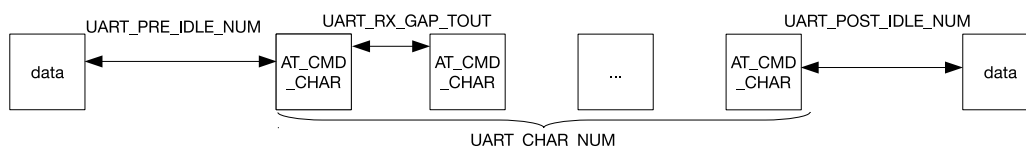


Figure 42.4-4. AT_CMD Character Structure

Figure 42.4-4 is the structure of a special character `AT_CMD`. If the receiver constantly receives `AT_CMD_CHAR` and the following conditions are met, a `UART_AT_CMD_CHAR_DET_INT` interrupt is generated.

- The interval between the first `AT_CMD_CHAR` and the last non-`AT_CMD_CHAR` character is at least `UART_PRE_IDLE_NUM` cycles.
- The interval between two `AT_CMD_CHAR` characters is less than `UART_RX_GAP_TOUT` in the unit of baud rate cycles.
- The number of `AT_CMD_CHAR` characters is equal to or greater than `UART_CHAR_NUM`.
- The interval between the last `AT_CMD_CHAR` character and next non-`AT_CMD_CHAR` character is at least `UART_POST_IDLE_NUM` cycles.

Note: Given that the interval between `AT_CMD_CHAR` characters is less than `UART_RX_GAP_TOUT` in the unit of baud rate cycles, the `APB_CLK` frequency is suggested not to be lower than 8 MHz.

42.4.6 RS485

The five regular UART controllers support RS485 communication mode. In this mode differential signals are used to transmit data, so it can communicate over longer distances at higher bit rates than RS232. RS485 has two-wire half-duplex and four-wire full-duplex options. UART controllers support two-wire half-duplex transmission and bus snooping.

42.4.6.1 Driver Control

As shown in Figure 42.4-5, in a two-wire multidrop network, an external RS485 transceiver is needed for differential to single-ended conversion or the other way around. An RS485 transceiver contains a driver and a receiver. When a UART controller is not in transmitter mode, the connection to the differential line can be broken by disabling the driver. When `DE` is 1, the driver is enabled; when `DE` is 0, the driver is disabled.

The UART receiver converts differential signals to single-ended signals via an external receiver. `RE` is the enable control signal for the receiver. When `RE` is 0, the receiver is enabled; when `RE` is 1, the receiver is disabled. If `RE` is configured as 0, the UART controller is allowed to snoop data on the bus, including the data sent by itself.

DE can be controlled by either software or hardware. To reduce the cost of software, in our design DE is controlled by hardware. As shown in Figure 42.4-5, DE is connected to dtrn_out of UART (please refer to Section 42.4.9.1 for more details).

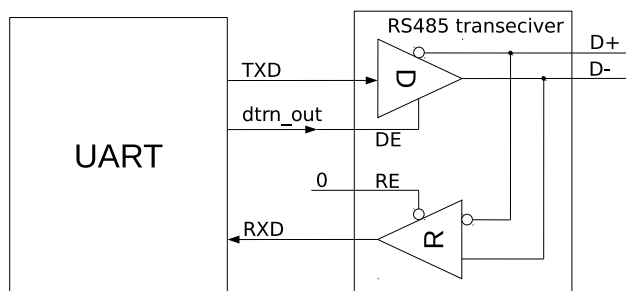


Figure 42.4-5. Driver Control Diagram in RS485 Mode

42.4.6.2 Turnaround Delay

By default, the UART controllers work in receiver mode. When a UART controller is switched from transmitter mode to receiver mode, the RS485 protocol requires a turnaround delay of one cycle after the stop bit. The UART transmitter supports adding a turnaround delay of one cycle before the start bit or after the stop bit. When `UART_DLO_EN` is set, a turnaround delay of one cycle is added before the start bit; when `UART_DL1_EN` is set, a turnaround delay of one cycle is added after the stop bit.

42.4.6.3 Bus Snooping

In a two-wire multidrop network, UART controllers support bus snooping if RE of the external RS485 transceiver is 0. By default, a UART controller is not allowed to transmit and receive data simultaneously. If `UART_RS485TX_RX_EN` is set and the external RS485 transceiver is configured as in Figure 42.4-5, a UART controller may receive data in transmitter mode and snoop the bus. If `UART_RS485RXBY_TX_EN` is set, a UART controller may transmit data in receiver mode.

The two UART controllers can snoop the data sent by themselves. In transmitter mode, when a UART controller monitors a collision between the data sent and the data received, a `UART_RS485_CLASH_INT` is generated; when a UART controller monitors a data frame error, a `UART_RS485_FRM_ERR_INT` interrupt is generated; when a UART controller monitors a parity bit error, a `UART_RS485_PARITY_ERR_INT` is generated.

42.4.7 IrDA

IrDA protocol consists of three layers, namely the physical layer, the link access protocol, and the link management protocol. The UART controllers implement IrDA's physical layer. In IrDA encoding, a UART controller supports data rates up to 115.2 kbit/s (SIR, or serial infrared mode). As shown in Figure 42.4-6, the IrDA encoder converts a non-return to zero code (NRZ) signal to a return to zero inverted code (RZI) signal and sends it to the external driver and infrared LED. This encoder uses modulated signals whose pulse width is 3/16 bits to indicate logic "0", and low levels to indicate logic "1". The IrDA decoder receives signals from the infrared receiver and converts them to NRZ signals. In most cases, the receiver is high when it is idle, and the parity bit of the encoder output is opposite to that of the decoder input. If a low pulse is detected, it indicates that a start bit has been received.

When IrDA function is enabled, one bit is divided into 16 clock cycles. If the bit to be sent is zero, then the 9th, 10th, and 11th clock cycle are high.

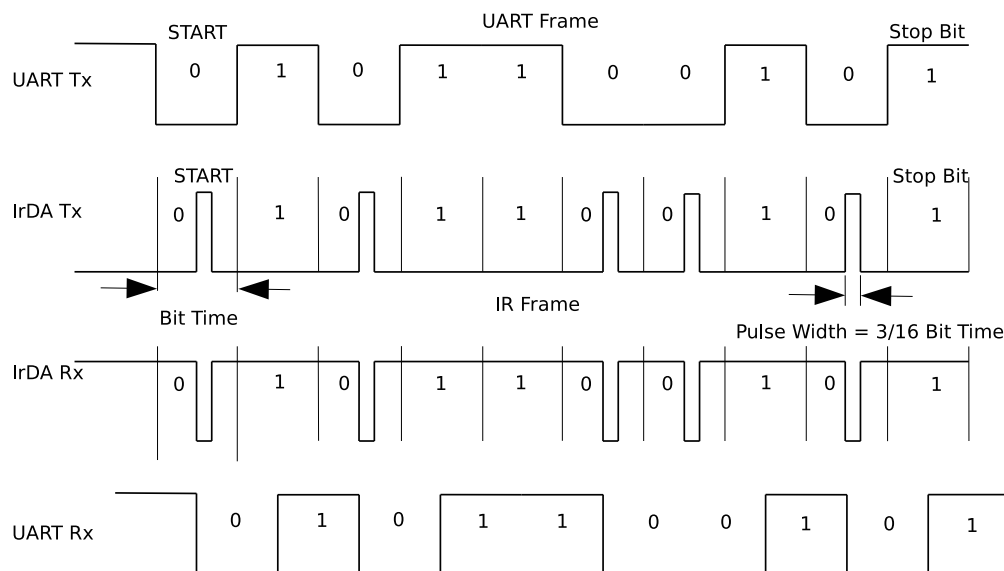


Figure 42.4-6. The Timing Diagram of Encoding and Decoding in SIR mode

The IrDA transceiver is half-duplex, meaning that it cannot send and receive data simultaneously. As shown in Figure 42.4-7, IrDA function is enabled by setting `UART_IRDA_EN`. When `UART_IRDA_TX_EN` is set to 1, the IrDA transceiver is enabled to send data and not allowed to receive data; when `UART_IRDA_TX_EN` is reset to 0, the IrDA transceiver is enabled to receive data and not allowed to send data.

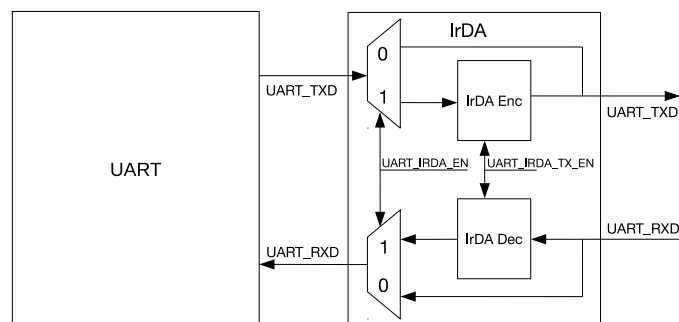


Figure 42.4-7. IrDA Encoding and Decoding Diagram

42.4.8 Wakeup

UART can be set as wakeup source. When a UART controller is in Light-sleep mode, a wake_up signal can be generated in four ways and be sent to the RTC module, which then wakes up ESP32-P4.

- `UART_WK_MODE_SEL` = 0: When all the clocks are disabled, the chip can be woken up by reverting RXD for multiple cycles until the number of positive edges is greater than or equal to `UART_ACTIVE_THRESHOLD` + 3.
- `UART_WK_MODE_SEL` = 1: UART Core keeps working, so the UART receiver can still receive data and store the received data in RX FIFO. When the number of data bytes in RX FIFO is greater than `UART_RX_WAKE_UP_THRHD`, the chip can be woken up from the Light-sleep mode.
- `UART_WK_MODE_SEL` = 2: When the UART receiver detects a start bit, the chip will be woken up.

- [UART_WK_MODE_SEL](#) = 3: When the UART receiver receives a specific character sequence, the chip will be woken up. The wakeup characters can be defined by configuring [UART_WK_CHAR0](#), [UART_WK_CHAR1](#), [UART_WK_CHAR2](#), [UART_WK_CHAR3](#), and [UART_WK_CHAR4](#). These characters can be formed into different character sequences by configuring [UART_CHAR_NUM](#) and [UART_WK_CHAR_MASK](#), as shown in Table 42.4-1. Once the sequence is detected, the chip will be woken up. For the last configuration in Table 42.4-1, UART will detect for CHAR0 ~ CHAR4 in order.

Table 42.4-1. UART_CHAR_WAKEUP Mode Configuration

UART_CHAR_NAME	UART_WP_CHAR_MASK	Character Sequence
1	0xF	CHAR4
2	0x7	CHAR3/CHAR4
3	0x3	CHAR2/CHAR3/CHAR4
4	0x1	CHAR1/CHAR2/CHAR3/CHAR4
5	0x0	CHAR0/CHAR1/CHAR2/CHAR3/CHAR4

After the chip is woken up by UART, it is necessary to clear the wake_up signal by transmitting data to UART in Active mode or resetting the whole UART, otherwise the number of rising edges required for the next wakeup will be reduced.

42.4.9 Flow Control

UART controllers have two ways to control data flow, namely hardware flow control and software flow control. Hardware flow control is achieved using output signal rtsn_out and input signal ctsn_in. Software flow control is achieved by inserting special characters in the data flow sent and detecting special characters in the data flow received.

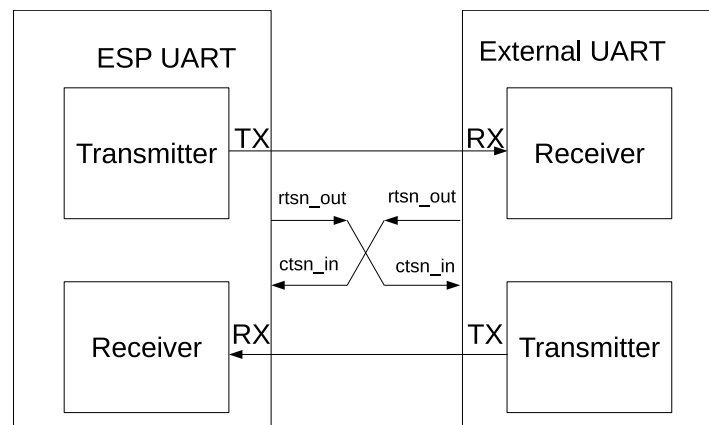


Figure 42.4-9. Connection between Hardware Flow Control Signals

When `ctsn_in` of IUO is low, IUO is allowed to send data; when `ctsn_in` is high, IUO is not allowed to send data. When IUO detects an edge change of `ctsn_in`, a `UART_CTS_CHG_INT` interrupt is generated.

If `dtrn_out` of IUO is high, it indicates that IUO is ready to transmit data. `dtrn_out` is generated by configuring the `UART_SW_DTR` field. When the IUO transmitter detects an edge change of `dtrn_in`, a `UART_DSR_CHG_INT` interrupt is generated. After this interrupt is detected, software can obtain the level of input signal `dtrn_in` by reading `UART_DSRN`. If `dtrn_in` is high, it indicates that EUO is ready to transmit data.

In a two-wire RS485 multidrop network enabled by setting `UART_RS485_EN`, `dtrn_out` is generated by hardware and used for transmit/receive turnaround. When data transmission starts, `dtrn_out` is pulled high and the external driver is enabled; when data transmission completes, `dtrn_out` is pulled low and the external driver is disabled. Please note that when there is a turnaround delay of one cycle added after the stop bit, `dtrn_out` is pulled low after the delay.

UART loopback test is enabled by setting `UART_LOOPBACK`. In the test, UART output signal `txd_out` is connected to its input signal `rx_d_in`, `rtsn_out` is connected to `ctsn_in`, and `dtrn_out` is connected to `dtrn_in`. If the data sent matches the data received, it indicates that UART controllers are working properly.

42.4.9.2 Software Flow Control

Instead of CTS/RTS lines, software flow control uses XON/XOFF characters to start or stop data transmission. Such flow control is enabled by setting `UART_SW_FLOW_CON_EN` to 1.

When using software flow control, hardware automatically detects if there are XON/XOFF characters in the data flow received, and generate a `UART_SW_XOFF_INT` or a `UART_SW_XON_INT` interrupt accordingly. If an XOFF character is detected, the transmitter stops data transmission once the current byte has been transmitted; if an XON character is detected, the transmitter starts data transmission. In addition, software can force the transmitter to stop sending data by setting `UART_FORCE_XOFF`, or to start sending data by setting `UART_FORCE_XON`.

Software determines whether to insert flow control characters based on the remaining room in RX FIFO. When `UART_SEND_XOFF` is set, the transmitter sends an XOFF character configured by `UART_XOFF_CHAR` after the current byte in transmission; when `UART_SEND_XON` is set, the transmitter sends an XON character configured by `UART_XON_CHAR` after the current byte in transmission. If the RX FIFO of a UART controller stores more data than `UART_XOFF_THRESHOLD`, `UART_SEND_XOFF` is set by hardware. As a result, the transmitter sends an XOFF character configured by `UART_XOFF_CHAR` after the current byte in transmission. If the RX FIFO of a

UART controller stores less data than `UART_XON_THRESHOLD`, `UART_SEND_XON` is set by hardware. As a result, the transmitter sends an XON character configured by `UART_XON_CHAR` after the current byte in transmission.

In full-duplex mode, when the UART receiver receives an XOFF character, the UART transmitter is not allowed to send any data including XOFF even if the UART receiver receives more data than its threshold. To avoid deadlocks in software flow control or overflow caused thereby, you can set `UART_XON_XOFF_STILL_SEND`. In this way, the UART transmitter can still send an XOFF character when it is not allowed to send any data.

42.4.10 GDMA Mode

The five UART controllers on ESP32-P4 share one TX/RX GDMA (General Direct Memory Access) channel via UHCI (Universal Host Controller Interface). In GDMA mode, UART controllers support the decoding and encoding of HCI data packets. The `UHCI_UART_SEL` field determines which UART controller occupies the GDMA TX/RX channel.

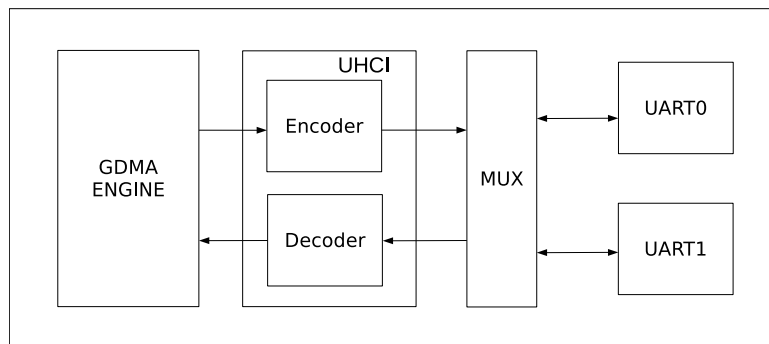


Figure 42.4-10. Data Transfer in GDMA Mode

Figure 42.4-10 shows how data is transferred using GDMA. Before GDMA receives data, software prepares an inlink. `GDMA_INLINK_ADDR_CH $n$` points to the first receive descriptor in the inlink. After `GDMA_INLINK_START_CH $n$` is set, UHCI sends data that UART has received to the decoder. The decoded data is then stored into the RAM pointed by the inlink under the control of GDMA.

Before GDMA sends data, software prepares an outlink and data to be sent. `GDMA_OUTLINK_ADDR_CH $n$` points to the first transmit descriptor in the outlink. After `GDMA_OUTLINK_START_CH $n$` is set, GDMA reads data from the RAM pointed by outlink. The data is then encoded by the encoder, and sent sequentially by the UART transmitter.

HCI data packets have separators at the beginning and the end, with data bits in the middle (separators + data bits + separators). The encoder inserts separators in front of and after data bits, and replaces data bits identical to separators with special characters. The decoder removes separators in front of and after data bits, and replaces special characters with separators. There can be more than one continuous separator at the beginning and the end of a data packet. The separator is configured by `UHCI_SEPER_CHAR`, 0xC0 by default. The special character is configured by `UHCI_ESC_SEQO_CHAR0` (0xDB by default) and `UHCI_ESC_SEQO_CHAR1` (0xDD by default). When all data has been sent, a `GDMA_OUT_TOTAL_EOF_CH $n$ _INT` interrupt is generated. When all data has been received, a `GDMA_IN_SUC_EOF_CH $n$ _INT` is generated.

42.4.11 Interrupts

ESP32-P4's UART_n and UHCI can generate the following interrupt signals that will be sent to the [Interrupt Matrix](#).

- UART_n_INTR
- UHCI_INTR

There are several internal interrupt sources from UART_n and UHCI that can generate the above interrupt signals.

UART_n interrupt sources are listed as follows:

- UART_AT_CMD_CHAR_DET_INT: Triggered when the receiver detects an AT_CMD character.
- UART_RS485_CLASH_INT: Triggered when a collision is detected between the transmitter and the receiver in RS485 mode.
- UART_RS485_FRM_ERR_INT: Triggered when an error is detected in the data frame sent by the transmitter in RS485 mode.
- UART_RS485_PARITY_ERR_INT: Triggered when an error is detected in the parity bit sent by the transmitter in RS485 mode.
- UART_TX_DONE_INT: Triggered when all data in the transmitter's TX FIFO has been sent.
- UART_TX_BRK_IDLE_DONE_INT: Triggered when the transmitter stays idle for the minimum interval (threshold) after sending the last data bit.
- UART_TX_BRK_DONE_INT: Triggered when the transmitter has sent all NULL characters after all data in TX FIFO had been sent.
- UART_GLITCH_DET_INT: Triggered when the receiver detects a glitch in the middle of the start bit.
- UART_SW_XOFF_INT: Triggered when [UART_SW_FLOW_CON_EN](#) is set and the receiver receives a XOFF character.
- UART_SW_XON_INT: Triggered when [UART_SW_FLOW_CON_EN](#) is set and the receiver receives a XON character.
- UART_RXFIFO_TOUT_INT: Triggered when the receiver has received at least one byte, and the bus remains idle for [UART_RX_TOUT_THRHD](#) bit time.
- UART_BRK_DET_INT: Triggered when the receiver detects a NULL character (i.e., logic 0 for one NULL character transmission) after stop bits.
- UART_CTS_CHG_INT: Triggered when the receiver detects an edge change of CTS_n signals.
- UART_DSR_CHG_INT: Triggered when the receiver detects an edge change of DSR_n signals.
- UART_RXFIFO_OVF_INT: Triggered when the amount of data received by the receiver exceeds the storage capacity of the FIFO.
- UART_FRM_ERR_INT: Triggered when the receiver detects a data frame error.
- UART_PARITY_ERR_INT: Triggered when the receiver detects a parity error.
- UART_TXFIFO_EMPTY_INT: Triggered when TX FIFO stores less data than what [UART_TXFIFO_EMPTY_THRHD](#) specifies.

- UART_RXFIFO_FULL_INT: Triggered when the receiver receives more data than what [UART_RXFIFO_FULL_THRHD](#) specifies.
- UART_WAKEUP_INT: Triggered when UART is woken up.

UHCI interrupt sources are listed as follows:

- UHCI_APP_CTRL1_INT: Triggered when software sets [UHCI_APP_CTRL1_INT_RAW](#).
- UHCI_APP_CTRL0_INT: Triggered when software sets [UHCI_APP_CTRL0_INT_RAW](#).
- UHCI_OUTLINK_EOF_ERR_INT: Triggered when an EOF error is detected in a transmit descriptor.
- UHCI_SEND_A_REG_Q_INT: Triggered when UHCI has sent a series of short packets using `always_send`.
- UHCI_SEND_S_REG_Q_INT: Triggered when UHCI has sent a series of short packets using `single_send`.
- UHCI_TX_HUNG_INT: Triggered when UHCI takes too long to read RAM using a GDMA transmit channel.
- UHCI_RX_HUNG_INT: Triggered when UHCI takes too long to receive data using a GDMA receive channel.
- UHCI_TX_START_INT: Triggered when GDMA detects a separator character.
- UHCI_RX_START_INT: Triggered when a separator character has been sent.

Each interrupt source can be configured by a common set of registers that are described in Section [Interrupt Configuration Registers](#). The specific registers can be found in Section [42.6 Register Summary](#).

42.5 Programming Procedures

42.5.1 Register Type

All UART registers are in the APB_CLK domain.

UART configuration registers can be classified into two groups. One group of registers are read in the APB_CLK or AHB_CLK domains, so once such registers are configured no extra operations are required. The other group of registers are read in the UART_SCLK domain, and therefore need to implement the clock domain crossing design. Once these registers are configured, the configured values need to be synchronized to the UART Core's clock domain by writing to [UART_REG_UPDATE](#). Once all values have been synchronized, [UART_REG_UPDATE](#) will be automatically cleared by hardware. After configuring registers that need synchronization, it is recommended to check whether [UART_REG_UPDATE](#) is 0. This is to ensure that register values configured before have already been synchronized.

To distinguish between these two groups of registers easily, all registers that implement the clock domain crossing design have the `_SYNC` suffix, and are put together in Section [42.6](#). Those without the `_SYNC` suffix in Section [42.6](#) are configuration registers that require no clock domain crossing.

42.5.2 Detailed Steps

Figure 42.5-1 illustrates the process to program UART controllers, namely initialize UART, configure registers, enable the UART transmitter or receiver, and finish data transmission.

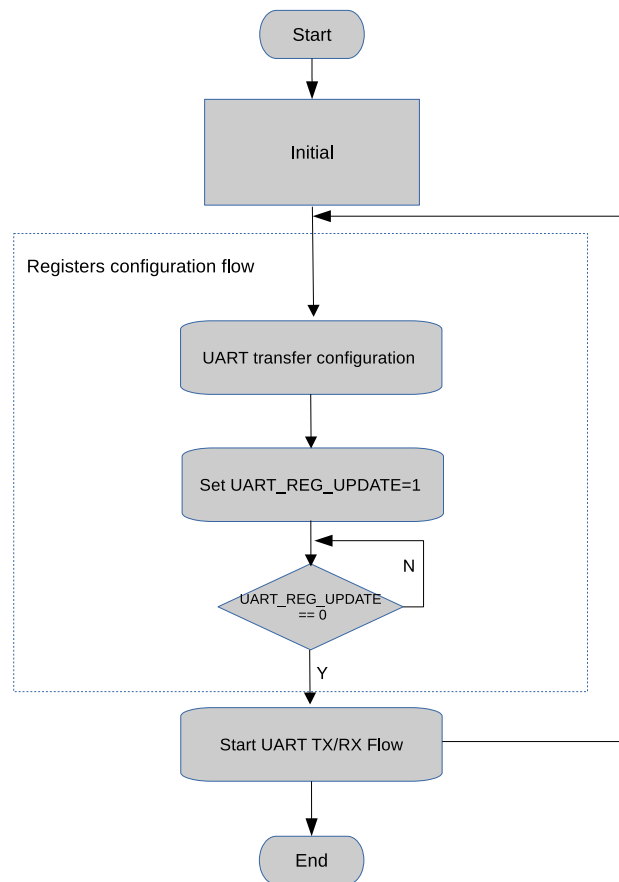


Figure 42.5-1. UART Programming Procedures

42.5.2.1 Initializing UART n

To initialize UART n :

- Write 1 to `HP_SYS_CLKRST_RST_EN_UART $n$ _APB`.
- Clear `HP_SYS_CLKRST_RST_EN_UART $n$ _APB` to 0.
- Write 1 to `HP_SYS_CLKRST_RST_EN_UART $n$ _CORE`.
- Clear `HP_SYS_CLKRST_RST_EN_UART $n$ _CORE` to 0.

42.5.2.2 Configuring UART n Communication

To configure UART n communication:

- Wait for `UART_REG_UPDATE` to become 0, which indicates the completion of the last synchronization.
- Select the clock source via `HP_SYS_CLKRST_UART $n$ _CLK_SRC_SEL`.
- Configure divisor of the divider via `HP_SYS_CLKRST_UART $n$ _SCLK_DIV_NUM`, `HP_SYS_CLKRST_UART $n$ _SCLK_DIV_DENOMINATOR`, and

[HP_SYS_CLKRST_UARTn_SCLK_DIV_NUMERATOR](#).

- Configure the baud rate for transmission via [UART_CLKDIV](#) and [UART_CLKDIV_FRAG](#).
- Configure data length via [UART_BIT_NUM](#).
- Configure odd or even parity check via [UART_PARITY_EN](#) and [UART_PARITY](#).
- Optional steps depending on application ...
- Synchronize the configured values to the Core Clock domain by writing 1 to [UART_REG_UPDATE](#).

42.5.2.3 Enabling UARTn

To enable UARTn transmitter:

- Configure TX FIFO's empty threshold via [UART_TXFIFO_EMPTY_THRHD](#).
- Disable UART_TXFIFO_EMPTY_INT interrupt by clearing [UART_TXFIFO_EMPTY_INT_ENA](#).
- Write data to be sent to [UART_RXFIFO_RD_BYTE](#).
- Clear UART_TXFIFO_EMPTY_INT interrupt by setting [UART_TXFIFO_EMPTY_INT_CLR](#).
- Enable UART_TXFIFO_EMPTY_INT interrupt by setting [UART_TXFIFO_EMPTY_INT_ENA](#).
- Check [UART_TXFIFO_EMPTY_INT_ST](#) and wait for the completion of data transmission.

To enable UARTn receiver:

- Configure RX FIFO's full threshold via [UART_RXFIFO_FULL_THRHD](#).
- Enable UART_RXFIFO_FULL_INT interrupt by setting [UART_RXFIFO_FULL_INT_ENA](#).
- Check [UART_RXFIFO_FULL_INT_ST](#) and wait until the RX FIFO is full.
- Read data from RX FIFO via [UART_RXFIFO_RD_BYTE](#), and obtain the number of bytes received in RX FIFO via [UART_RXFIFO_CNT](#).

42.6 Register Summary

42.6.1 UART Register Summary

The addresses in this section are relative to UART Controller base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
FIFO Configuration			
UART_FIFO_REG	FIFO data register	0x0000	RO
UART_TOUT_CONF_SYNC_REG	UART threshold and allocation configuration	0x0064	R/W
UART Interrupt Register			
UART_INT_RAW_REG	Raw interrupt status	0x0004	R/WTC/SS
UART_INT_ST_REG	Masked interrupt status	0x0008	RO
UART_INT_ENA_REG	Interrupt enable bits	0x000C	R/W
UART_INT_CLR_REG	Interrupt clear bits	0x0010	WT
Configuration Register			
UART_CLKDIV_SYNC_REG	Clock divider configuration	0x0014	R/W
UART_RX_FILT_REG	RX filter configuration	0x0018	R/W
UART_CONFO_SYNC_REG	Configuration register 0	0x0020	R/W
UART_CONF1_REG	Configuration register 1	0x0024	R/W
UART_HWFC_CONF_SYNC_REG	Hardware flow control configuration	0x002C	R/W
UART_SLEEP_CONFO_REG	UART sleep configuration register 0	0x0030	R/W
UART_SLEEP_CONF1_REG	UART sleep configuration register 1	0x0034	R/W
UART_SLEEP_CONF2_REG	UART sleep configuration register 2	0x0038	R/W
UART_SWFC_CONFO_SYNC_REG	Software flow control character configuration	0x003C	varies
UART_SWFC_CONF1_REG	Software flow control character configuration	0x0040	R/W
UART_TXBRK_CONF_SYNC_REG	TX break character configuration	0x0044	R/W
UART_IDLE_CONF_SYNC_REG	Frame end idle time configuration	0x0048	R/W
UART_RS485_CONF_SYNC_REG	RS485 mode configuration	0x004C	R/W
UART_CLK_CONF_REG	UART core clock configuration	0x0088	R/W
UART_REG_UPDATE_REG	UART register configuration update	0x0098	R/W/SC
UART_ID_REG	UART ID register	0x009C	R/W
Status Register			
UART_STATUS_REG	UART status register	0x001C	RO
UART_MEM_TX_STATUS_REG	TX FIFO write and read offset address	0x0068	RO
UART_MEM_RX_STATUS_REG	Rx FIFO write and read offset address	0x006C	RO
UART_FSM_STATUS_REG	UART transmit and receive status	0x0070	RO
UART_AFIFO_STATUS_REG	UART asynchronous FIFO status	0x0090	RO
AT Escape Sequence Selection Configuration			
UART_AT_CMD_PRECNT_SYNC_REG	Pre-sequence timing configuration	0x0050	R/W
UART_AT_CMD_POSTCNT_SYNC_REG	Post-sequence timing configuration	0x0054	R/W
UART_AT_CMD_GAPTOUT_SYNC_REG	Timeout configuration	0x0058	R/W

Name	Description	Address	Access
UART_AT_CMD_CHAR_SYNC_REG	AT escape sequence detection configuration	0x005C	R/W
Autobaud Register			
UART_POSPULSE_REG	Autobaud high pulse register	0x0074	RO
UART_NEGPULSE_REG	Autobaud low pulse register	0x0078	RO
UART_LOWPULSE_REG	Autobaud minimum low pulse duration register	0x007C	RO
UART_HIGHPULSE_REG	Autobaud minimum high pulse duration register	0x0080	RO
UART_RXD_CNT_REG	Autobaud edge change count register	0x0084	RO
Version Register			
UART_DATE_REG	UART version control register	0x008C	R/W

42.6.2 LP UART Register Summary

The addresses in this section are relative to LP UART base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
FIFO Configuration			
LP_UART_FIFO_REG	FIFO data register	0x0000	RO
LP_UART_TOUT_CONF_SYNC_REG	LP UART threshold and allocation configuration	0x0064	R/W
LP UART Interrupt Register			
LP_UART_INT_RAW_REG	Raw interrupt status	0x0004	R/WTC/SS
LP_UART_INT_ST_REG	Masked interrupt status	0x0008	RO
LP_UART_INT_ENA_REG	Interrupt enable bits	0x000C	R/W
LP_UART_INT_CLR_REG	Interrupt clear bits	0x0010	WT
Configuration Register			
LP_UART_CLKDIV_SYNC_REG	Clock divider configuration	0x0014	R/W
LP_UART_RX_FILT_REG	RX filter configuration	0x0018	R/W
LP_UART_CONFO_SYNC_REG	Configuration register 0	0x0020	R/W
LP_UART_CONF1_REG	Configuration register 1	0x0024	R/W
LP_UART_HWFC_CONF_SYNC_REG	Hardware flow control configuration	0x002C	R/W
LP_UART_SLEEP_CONFO_REG	LP UART sleep configuration register 0	0x0030	R/W
LP_UART_SLEEP_CONF1_REG	LP UART sleep configuration register 1	0x0034	R/W
LP_UART_SLEEP_CONF2_REG	LP UART sleep configuration register 2	0x0038	R/W
LP_UART_SWFC_CONFO_SYNC_REG	Software flow control character configuration	0x003C	varies
LP_UART_SWFC_CONF1_REG	Software flow control character configuration	0x0040	R/W
LP_UART_TXBRK_CONF_SYNC_REG	TX break character configuration	0x0044	R/W
LP_UART_IDLE_CONF_SYNC_REG	Frame end idle time configuration	0x0048	R/W
LP_UART_CLK_CONF_REG	LP UART core clock configuration	0x0088	R/W

Name	Description	Address	Access
LP_UART_REG_UPDATE_REG	LP UART register configuration update register	0x0098	R/W/SC
LP_UART_ID_REG	LP UART ID register	0x009C	R/W
Status Register			
LP_UART_STATUS_REG	LP UART status register	0x001C	RO
LP_UART_MEM_TX_STATUS_REG	TX FIFO write and read offset address	0x0068	RO
LP_UART_MEM_RX_STATUS_REG	RX FIFO write and read offset address	0x006C	RO
LP_UART_FSM_STATUS_REG	LP UART transmit and receive status	0x0070	RO
LP_UART_AFIFO_STATUS_REG	LP UART asynchronous FIFO Status	0x0090	RO
AT Escape Sequence Selection Configuration			
LP_UART_AT_CMD_PRECNT_SYNC_REG	Pre-sequence timing configuration	0x0050	R/W
LP_UART_AT_CMD_POSTCNT_SYNC_REG	Post-sequence timing configuration	0x0054	R/W
LP_UART_AT_CMD_GAPTOOUT_SYNC_REG	Timeout configuration	0x0058	R/W
LP_UART_AT_CMD_CHAR_SYNC_REG	AT escape sequence detection configuration	0x005C	R/W
Version Register			
LP_UART_DATE_REG	LP UART version register	0x008C	R/W

42.6.3 UHCI Register Summary

The addresses in this section are relative to UHCI base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Configuration Register			
UHCI_CONFO_REG	UHCI configuration register	0x0000	R/W
UHCI_CONF1_REG	UHCI configuration register	0x0014	varies
UHCI_ESCAPE_CONF_REG	Escape character configuration	0x0020	R/W
UHCI_HUNG_CONF_REG	Timeout configuration	0x0024	R/W
UHCI_ACK_NUM_REG	UHCI ACK number configuration	0x0028	varies
UHCI_QUICK_SENT_REG	UHCI quick send configuration register	0x0030	varies
UHCI_REG_Q0_WORD0_REG	Q0 WORD0 quick send register	0x0034	R/W
UHCI_REG_Q0_WORD1_REG	Q0 WORD1 quick send register	0x0038	R/W
UHCI_REG_Q1_WORD0_REG	Q1 WORD0 quick send register	0x003C	R/W
UHCI_REG_Q1_WORD1_REG	Q1 WORD1 quick send register	0x0040	R/W
UHCI_REG_Q2_WORD0_REG	Q2 WORD0 quick send register	0x0044	R/W
UHCI_REG_Q2_WORD1_REG	Q2 WORD1 quick send register	0x0048	R/W
UHCI_REG_Q3_WORD0_REG	Q3 WORD0 quick send register	0x004C	R/W
UHCI_REG_Q3_WORD1_REG	Q3 WORD1 quick send register	0x0050	R/W
UHCI_REG_Q4_WORD0_REG	Q4 WORD0 quick send register	0x0054	R/W
UHCI_REG_Q4_WORD1_REG	Q4 WORD1 quick send register	0x0058	R/W
UHCI_REG_Q5_WORD0_REG	Q5 WORD0 quick send register	0x005C	R/W

Name	Description	Address	Access
UHCI_REG_Q5_WORD1_REG	Q5 WORD1 quick send register	0x0060	R/W
UHCI_REG_Q6_WORD0_REG	Q6 WORD0 quick send register	0x0064	R/W
UHCI_REG_Q6_WORD1_REG	Q6 WORD1 quick register	0x0068	R/W
UHCI_ESC_CONF0_REG	Escape sequence configuration register 0	0x006C	R/W
UHCI_ESC_CONF1_REG	Escape sequence configuration register 1	0x0070	R/W
UHCI_ESC_CONF2_REG	Escape sequence configuration register 2	0x0074	R/W
UHCI_ESC_CONF3_REG	Escape sequence configuration register 3	0x0078	R/W
UHCI_PKT_THRES_REG	Configuration register for packet length	0x007C	R/W
UHCI Interrupt Register			
UHCI_INT_RAW_REG	Raw interrupt status	0x0004	varies
UHCI_INT_ST_REG	Masked interrupt status	0x0008	RO
UHCI_INT_ENA_REG	Interrupt enable bits	0x000C	R/W
UHCI_INT_CLR_REG	Interrupt clear bits	0x0010	WT
UHCI Status Register			
UHCI_STATE0_REG	UHCI receive status	0x0018	RO
UHCI_STATE1_REG	UHCI transmit status	0x001C	RO
UHCI_RX_HEAD_REG	UHCI packet header register	0x002C	RO
Version Register			
UHCI_DATE_REG	UHCI version control register	0x0080	R/W

42.7 Registers

42.7.1 UART Registers

The addresses in this section are relative to UART Controller base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

Register 42.1. UART_FIFO_REG (0x0000)

(reserved)																										UART_RXFIFO_RD_BYTE									
31																										8	7	0							
0 0																												0							Reset

UART_RXFIFO_RD_BYTE Represents the data UART *n* read from FIFO.
Measurement unit: byte. (RO)

Register 42.2. UART_TOUT_CONF_SYNC_REG (0x0064)

(reserved)																				UART_RX_TOUT_THRHD												UART_RX_TOUT_FLOW_DIS		UART_RX_TOUT_EN													
31																				12												11												2		1	0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																				0xa												0		0	Reset												

UART_RX_TOUT_EN Configures whether or not to enable UART receiver's timeout function.

0: Disable

1: Enable

(R/W)

UART_RX_TOUT_FLOW_DIS Configures whether or not to disable the idle status counter when hardware flow control is enabled.

0: Enable

1: Disable

(R/W)

UART_RX_TOUT_THRHD Configures the amount of time that the bus can remain idle before timeout.

Measurement unit: bit time (the time to transmit 1 bit). (R/W)

Register 42.3. UART_INT_RAW_REG (0x0004)

(reserved)																UART_WAKEUP_INT_RAW UART_AT_CMD_CHAR_DET_INT_RAW UART_RS485_CLASH_INT_RAW UART_RS485_FRM_ERR_INT_RAW UART_RS485_Parity_ERR_INT_RAW UART_TX_DONE_INT_RAW UART_TX_BRK_IDLE_DONE_INT_RAW UART_GLITCH_DET_INT_RAW UART_SW_XON_INT_RAW UART_SW_XOFF_INT_RAW UART_RXFIFO_TOUT_INT_RAW UART_CTS_CHG_INT_RAW UART_DSR_CHG_INT_RAW UART_FRM_ERR_INT_RAW UART_PARITY_ERR_INT_RAW UART_TXFIFO_EMPTY_INT_RAW UART_RXFIFO_FULL_INT_RAW																			
31													20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	Reset						

UART_RXFIFO_FULL_INT_RAW The raw interrupt status of UART_RXFIFO_FULL_INT. (R/WTC/SS)

UART_TXFIFO_EMPTY_INT_RAW The raw interrupt status of UART_TXFIFO_EMPTY_INT. (R/WTC/SS)

UART_PARITY_ERR_INT_RAW The raw interrupt status of UART_PARITY_ERR_INT. (R/WTC/SS)

UART_FRM_ERR_INT_RAW The raw interrupt status of UART_FRM_ERR_INT. (R/WTC/SS)

UART_RXFIFO_OVF_INT_RAW The raw interrupt status of UART_RXFIFO_OVF_INT. (R/WTC/SS)

UART_DSR_CHG_INT_RAW The raw interrupt status of UART_DSR_CHG_INT. (R/WTC/SS)

UART_CTS_CHG_INT_RAW The raw interrupt status of UART_CTS_CHG_INT. (R/WTC/SS)

UART_BRK_DET_INT_RAW The raw interrupt status of UART_BRK_DET_INT. (R/WTC/SS)

UART_RXFIFO_TOUT_INT_RAW The raw interrupt status of UART_RXFIFO_TOUT_INT. (R/WTC/SS)

UART_SW_XON_INT_RAW The raw interrupt status of UART_SW_XON_INT. (R/WTC/SS)

UART_SW_XOFF_INT_RAW UART_SW_XOFF_INT. (R/WTC/SS)

UART_GLITCH_DET_INT_RAW The raw interrupt status of UART_GLITCH_DET_INT. (R/WTC/SS)

UART_TX_BRK_DONE_INT_RAW The raw interrupt status of UART_TX_BRK_DONE_INT. (R/WTC/SS)

UART_TX_BRK_IDLE_DONE_INT_RAW The raw interrupt status of UART_TX_BRK_IDLE_DONE_INT.
(R/WTC/SS)

UART_TX_DONE_INT_RAW The raw interrupt status of UART_TX_DONE_INT. (R/WTC/SS)

UART_RS485_PARITY_ERR_INT_RAW The raw interrupt status of UART_RS485_PARITY_ERR_INT.
(R/WTC/SS)

UART_RS485_FRM_ERR_INT_RAW The raw interrupt status of UART_RS485_FRM_ERR_INT.
(R/WTC/SS)

Continued on the next page...

Register 42.3. UART_INT_RAW_REG (0x0004)

Continued from the previous page...

UART_RS485_CLASH_INT_RAW The raw interrupt status of UART_RS485_CLASH_INT. (R/WTC/SS)

UART_AT_CMD_CHAR_DET_INT_RAW The raw interrupt status of UART_AT_CMD_CHAR_DET_INT.
(R/WTC/SS)

UART_WAKEUP_INT_RAW The raw interrupt status of UART_WAKEUP_INT. (R/WTC/SS)

Register 42.4. UART_INT_ST_REG (0x0008)

(reserved)																															UART_WAKEUP_INT_ST UART_AT_CMD_CHAR_DET_INT_ST UART_RS485_CLASH_INT_ST UART_RS485_FRM_ERR_INT_ST UART_RS485_PARITY_ERR_INT_ST UART_TX_DONE_INT_ST UART_TX_BRK_IDLE_DONE_INT_ST UART_GLITCH_DET_INT_ST UART_SW_XON_INT_ST UART_SW_XOFF_INT_ST UART_RXFIFO_TOUT_INT_ST UART_BRK_DET_INT_ST UART_CTS_CHG_INT_ST UART_DSR_CHG_INT_ST UART_RXFIFO_OVF_INT_ST UART_FRM_ERR_INT_ST UART_PARITY_ERR_INT_ST UART_TXFIFO_EMPTY_INT_ST UART_RXFIFO_FULL_INT_ST															
31																								20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset										

UART_RXFIFO_FULL_INT_ST The masked interrupt status of UART_RXFIFO_FULL_INT. (RO)

UART_TXFIFO_EMPTY_INT_ST The masked interrupt status of UART_TXFIFO_EMPTY_INT. (RO)

UART_PARITY_ERR_INT_ST The masked interrupt status of UART_PARITY_ERR_INT. (RO)

UART_FRM_ERR_INT_ST The masked interrupt status of UART_FRM_ERR_INT. (RO)

UART_RXFIFO_OVF_INT_ST The masked interrupt status of UART_RXFIFO_OVF_INT. (RO)

UART_DSR_CHG_INT_ST The masked interrupt status of UART_DSR_CHG_INT. (RO)

UART_CTS_CHG_INT_ST The masked interrupt status of UART_CTS_CHG_INT. (RO)

UART_BRK_DET_INT_ST The masked interrupt status of UART_BRK_DET_INT. (RO)

UART_RXFIFO_TOUT_INT_ST The masked interrupt status of UART_RXFIFO_TOUT_INT. (RO)

UART_SW_XON_INT_ST The masked interrupt status of UART_SW_XON_INT. (RO)

UART_SW_XOFF_INT_ST The masked interrupt status of UART_SW_XOFF_INT. (RO)

UART_GLITCH_DET_INT_ST The masked interrupt status of UART_GLITCH_DET_INT. (RO)

UART_TX_BRK_DONE_INT_ST The masked interrupt status of UART_TX_BRK_DONE_INT. (RO)

UART_TX_BRK_IDLE_DONE_INT_ST The masked interrupt status of UART_TX_BRK_IDLE_DONE_INT. (RO)

UART_TX_DONE_INT_ST The masked interrupt status of UART_TX_DONE_INT. (RO)

UART_RS485_PARITY_ERR_INT_ST The masked interrupt status of UART_RS485_PARITY_ERR_INT. (RO)

UART_RS485_FRM_ERR_INT_ST The masked interrupt status of UART_RS485_FRM_ERR_INT. (RO)

UART_RS485_CLASH_INT_ST The masked interrupt status of UART_RS485_CLASH_INT. (RO)

UART_AT_CMD_CHAR_DET_INT_ST The masked interrupt status of UART_AT_CMD_CHAR_DET_INT. (RO)

UART_WAKEUP_INT_ST The masked interrupt status of UART_WAKEUP_INT. (RO)

Register 42.5. UART_INT_ENA_REG (0x000C)

(reserved)																UART_WAKEUP_INT_ENA UART_AT_CMD_CHAR_DET_INT_ENA UART_RS485_CLASH_INT_ENA UART_RS485_FRM_ERR_INT_ENA UART_RS485_PARITY_ERR_INT_ENA UART_TX_DONE_INT_ENA UART_TX_BRK_IDLE_DONE_INT_ENA UART_GLITCH_DET_INT_ENA UART_SW_XOFF_INT_ENA UART_SW_XON_INT_ENA UART_RXFIFO_TOUT_INT_ENA UART_BRK_DET_INT_ENA UART_CTS_CHG_INT_ENA UART_DSR_CHG_INT_ENA UART_RXFIFO_OVF_INT_ENA UART_FRM_ERR_INT_ENA UART_PARITY_ERR_INT_ENA UART_TXFIFO_EMPTY_INT_ENA UART_RXFIFO_FULL_INT_ENA																			
31													20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset					

UART_RXFIFO_FULL_INT_ENA Write 1 to enable UART_RXFIFO_FULL_INT. (R/W)

UART_TXFIFO_EMPTY_INT_ENA Write 1 to enable UART_TXFIFO_EMPTY_INT. (R/W)

UART_PARITY_ERR_INT_ENA Write 1 to enable UART_PARITY_ERR_INT. (R/W)

UART_FRM_ERR_INT_ENA Write 1 to enable UART_FRM_ERR_INT. (R/W)

UART_RXFIFO_OVF_INT_ENA Write 1 to enable UART_RXFIFO_OVF_INT. (R/W)

UART_DSR_CHG_INT_ENA Write 1 to enable UART_DSR_CHG_INT. (R/W)

UART_CTS_CHG_INT_ENA Write 1 to enable UART_CTS_CHG_INT. (R/W)

UART_BRK_DET_INT_ENA Write 1 to enable UART_BRK_DET_INT. (R/W)

UART_RXFIFO_TOUT_INT_ENA Write 1 to enable UART_RXFIFO_TOUT_INT. (R/W)

UART_SW_XON_INT_ENA Write 1 to enable UART_SW_XON_INT. (R/W)

UART_SW_XOFF_INT_ENA Write 1 to enable UART_SW_XOFF_INT. (R/W)

UART_GLITCH_DET_INT_ENA Write 1 to enable UART_GLITCH_DET_INT. (R/W)

UART_TX_BRK_DONE_INT_ENA Write 1 to enable UART_TX_BRK_DONE_INT. (R/W)

UART_TX_BRK_IDLE_DONE_INT_ENA Write 1 to enable UART_TX_BRK_IDLE_DONE_INT. (R/W)

UART_TX_DONE_INT_ENA Write 1 to enable UART_TX_DONE_INT. (R/W)

UART_RS485_PARITY_ERR_INT_ENA Write 1 to enable UART_RS485_PARITY_ERR_INT. (R/W)

UART_RS485_FRM_ERR_INT_ENA Write 1 to enable UART_RS485_FRM_ERR_INT. (R/W)

UART_RS485_CLASH_INT_ENA Write 1 to enable UART_RS485_CLASH_INT. (R/W)

UART_AT_CMD_CHAR_DET_INT_ENA Write 1 to enable UART_AT_CMD_CHAR_DET_INT. (R/W)

UART_WAKEUP_INT_ENA Write 1 to enable UART_WAKEUP_INT. (R/W)

Register 42.6. UART_INT_CLR_REG (0x0010)

(reserved)																UART_WAKEUP_INT_CLR UART_AT_CMD_CHAR_DET_INT_CLR UART_RS485_CLASH_INT_CLR UART_RS485_FRM_ERR_INT_CLR UART_RS485_PARITY_ERR_INT_CLR UART_TX_DONE_INT_CLR UART_TX_BRK_IDLE_DONE_INT_CLR UART_GLITCH_DET_INT_CLR UART_SW_XOFF_DET_INT_CLR UART_SW_XON_INT_CLR UART_RXFIFO_TOUT_INT_CLR UART_BRK_DET_INT_CLR UART_CTS_CHG_INT_CLR UART_DSR_CHG_INT_CLR UART_RXFIFO_OVF_INT_CLR UART_FRM_ERR_INT_CLR UART_PARITY_ERR_INT_CLR UART_TXFIFO_EMPTY_INT_CLR UART_RXFIFO_FULL_INT_CLR																
31												20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset		

UART_RXFIFO_FULL_INT_CLR Write 1 to clear UART_RXFIFO_FULL_INT. (WT)

UART_TXFIFO_EMPTY_INT_CLR Write 1 to clear UART_TXFIFO_EMPTY_INT. (WT)

UART_PARITY_ERR_INT_CLR Write 1 to clear UART_PARITY_ERR_INT. (WT)

UART_FRM_ERR_INT_CLR Write 1 to clear UART_FRM_ERR_INT. (WT)

UART_RXFIFO_OVF_INT_CLR Write 1 to clear UART_RXFIFO_OVF_INT. (WT)

UART_DSR_CHG_INT_CLR Write 1 to clear UART_DSR_CHG_INT. (WT)

UART_CTS_CHG_INT_CLR Write 1 to clear UART_CTS_CHG_INT. (WT)

UART_BRK_DET_INT_CLR Write 1 to clear UART_BRK_DET_INT. (WT)

UART_RXFIFO_TOUT_INT_CLR Write 1 to clear UART_RXFIFO_TOUT_INT. (WT)

UART_SW_XON_INT_CLR Write 1 to clear UART_SW_XON_INT. (WT)

UART_SW_XOFF_INT_CLR Write 1 to clear UART_SW_XOFF_INT. (WT)

UART_GLITCH_DET_INT_CLR Write 1 to clear UART_GLITCH_DET_INT. (WT)

UART_TX_BRK_DONE_INT_CLR Write 1 to clear UART_TX_BRK_DONE_INT. (WT)

UART_TX_BRK_IDLE_DONE_INT_CLR Write 1 to clear UART_TX_BRK_IDLE_DONE_INT. (WT)

UART_TX_DONE_INT_CLR Write 1 to clear UART_TX_DONE_INT. (WT)

UART_RS485_PARITY_ERR_INT_CLR Write 1 to clear UART_RS485_PARITY_ERR_INT. (WT)

UART_RS485_FRM_ERR_INT_CLR Write 1 to clear UART_RS485_FRM_ERR_INT. (WT)

UART_RS485_CLASH_INT_CLR Write 1 to clear UART_RS485_CLASH_INT. (WT)

UART_AT_CMD_CHAR_DET_INT_CLR Write 1 to clear UART_AT_CMD_CHAR_DET_INT. (WT)

UART_WAKEUP_INT_CLR Write 1 to clear UART_WAKEUP_INT. (WT)

Register 42.7. UART_CLKDIV_SYNC_REG (0x0014)

(reserved)								UART_CLKDIV_FRAG												(reserved)								UART_CLKDIV																																																																																																			
31								24								23								20								19								12								11								0																																																																							
0								0								0								0								0								0								0								0								0x0								0								0								0								0								0								0x2b6								Reset							

UART_CLKDIV Configures the integral part of the divisor for baud rate generation. (R/W)

UART_CLKDIV_FRAG Configures the fractional part of the divisor for baud rate generation. (R/W)

Register 42.8. UART_RX_FILT_REG (0x0018)

(reserved)																UART_GLITCH_FILT_EN		UART_GLITCH_FILT																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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UART_GLITCH_FILT Configures the width of a pulse to be filtered.

Measurement unit: UART Core's clock cycle.

Pulses whose width is lower than this value will be ignored. (R/W)

UART_GLITCH_FILT_EN Configures whether or not to enable RX signal filter.

0: Disable

1: Enable

(R/W)

Register 42.9. UART_CONFO_SYNC_REG (0x0020)

(reserved)								UART_TXFIFO_RST UART_RXFIFO_RST UART_SW_RTS UART_MEM_RTS UART_AUTOBAUD_EN UART_ERR_WD_MASK UART_TXD_INV UART_RXD_INV UART_IRDA_EN UART_TX_FLOW_EN UART_LOOPBACK UART_IRDA_RX_INV UART_IRDA_TX_INV UART_IRDA_WCTL UART_IRDA_TX_EN UART_TXD_DPLX UART_STOP_BIT_NUM UART_BIT_NUM UART_PARITY_EN UART_PARITY																		
31	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	3	0	0			

UART_PARITY Configures the parity check mode.

0: Even parity

1: Odd parity

(R/W)

UART_PARITY_EN Configures whether or not to enable UART parity check.

0: Disable

1: Enable

(R/W)

UART_BIT_NUM Configures the number of data bits.

0: 5 bits

1: 6 bits

2: 7 bits

3: 8 bits

(R/W)

UART_STOP_BIT_NUM Configures the number of stop bits.

0: Invalid. No effect

1: 1 bit

2: 1.5 bits

3: 2 bits

(R/W)

UART_TXD_BRK Configures whether or not to send NULL characters when finishing data transmission.

0: Not send

1: Send

(R/W)

UART_IRDA_DPLX Configures whether or not to enable IrDA loopback test.

0: Disable

1: Enable

(R/W)

UART_IRDA_TX_EN Configures whether or not to enable the IrDA transmitter.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 42.9. UART_CONFO_SYNC_REG (0x0020)

Continued from the previous page...

UART_IRDA_WCTL Configures the 11th bit of the IrDA transmitter.

0: This bit is 0.

1: This bit is the same as the 10th bit.

(R/W)

UART_IRDA_TX_INV Configures whether or not to invert the level of the IrDA transmitter.

0: Not invert

1: Invert

(R/W)

UART_IRDA_RX_INV Configures whether or not to invert the level of the IrDA receiver.

0: Not invert

1: Invert

(R/W)

UART_LOOPBACK Configures whether or not to enable UART loopback test.

0: Disable

1: Enable

(R/W)

UART_TX_FLOW_EN Configures whether or not to enable flow control for the transmitter.

0: Disable

1: Enable

(R/W)

UART_IRDA_EN Configures whether or not to enable IrDA protocol.

0: Disable

1: Enable

(R/W)

UART_RXD_INV Configures whether or not to invert the level of UART RXD signal.

0: Not invert

1: Invert

(R/W)

UART_TXD_INV Configures whether or not to invert the level of UART TXD signal.

0: Not invert

1: Invert

(R/W)

UART_DIS_RX_DAT_OVF Configures whether or not to disable data overflow detection for the UART receiver.

0: Enable

1: Disable

(R/W)

Continued on the next page...

Register 42.9. UART_CONFO_SYNC_REG (0x0020)

Continued from the previous page...

UART_ERR_WR_MASK Configures whether or not to store the received data with errors into FIFO.

- 0: Store
 - 1: Not store
- (R/W)

UART_AUTOBAUD_EN Configures whether or not to enable baud rate detection.

- 0: Disable
 - 1: Enable
- (R/W)

UART_MEM_CLK_EN Configures whether or not to enable clock gating for UART memory.

- 0: Disable
 - 1: Enable
- (R/W)

UART_SW_RTS Configures the RTS signal used in software flow control.

- 0: The UART transmitter is not allowed to send data.
 - 1: The UART transmitted is allowed to send data.
- (R/W)

UART_RXFIFO_RST Configures whether or not to reset the UART RX FIFO.

- 0: Not reset
 - 1: Reset
- (R/W)

UART_TXFIFO_RST Configures whether or not to reset the UART TX FIFO.

- 0: Not reset
 - 1: Reset
- (R/W)

Register 42.10. UART_CONF1_REG (0x0024)

(reserved)										UART_CLK_EN UART_SW_DTR UART_DTR_INV UART_RTS_INV UART_DSR_INV UART_CTS_INV						UART_TXFIFO_EMPTY_THRHD								UART_RXFIFO_FULL_THRHD													
31											22	21	20	19	18	17	16	15									8	7									0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x60								0x60								Reset					

UART_RXFIFO_FULL_THRHD Configures the threshold for RX FIFO being full.
Measurement unit: byte. (R/W)

UART_TXFIFO_EMPTY_THRHD Configures the threshold for TX FIFO being empty.
Measurement unit: byte. (R/W)

UART_CTS_INV Configures whether or not to invert the level of UART CTS signal.
0: Not invert
1: Invert
(R/W)

UART_DSR_INV Configures whether or not to invert the level of UART DSR signal.
0: Not invert
1: Invert
(R/W)

UART_RTS_INV Configures whether or not to invert the level of UART RTS signal.
0: Not invert
1: Invert
(R/W)

UART_DTR_INV Configures whether or not to invert the level of UART DTR signal.
0: Not invert
1: Invert
(R/W)

UART_SW_DTR Configures the DTR signal used in software flow control.
0: Data to be transmitted is not ready.
1: Data to be transmitted is ready.
(R/W)

UART_CLK_EN Configures clock gating.
0: Support clock only when the application writes registers.
1: Always force the clock on for registers.
(R/W)

Register 42.11. UART_HWFC_CONF_SYNC_REG (0x002C)

(reserved)																UART_RX_FLOW_EN		UART_RX_FLOW_THRHD	
31																	8	7	0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0x0		Reset

UART_RX_FLOW_THRHD Configures the maximum number of data bytes that can be received during hardware flow control.

Measurement unit: byte. (R/W)

UART_RX_FLOW_EN Configures whether or not to enable the UART receiver.

0: Disable

1: Enable

(R/W)

Register 42.12. UART_SLEEP_CONFO_REG (0x0030)

UART_WK_CHAR4								UART_WK_CHAR3								UART_WK_CHAR2								UART_WK_CHAR1							
31				24				23				16				15				8				7				0			
0x0				0x0				0x0				0x0				0x0				0x0				Reset							

UART_WK_CHAR1 Configures wakeup character 1. (R/W)

UART_WK_CHAR2 Configures wakeup character 2. (R/W)

UART_WK_CHAR3 Configures wakeup character 3. (R/W)

UART_WK_CHAR4 Configures wakeup character 4. (R/W)

Register 42.13. UART_SLEEP_CONF1_REG (0x0034)

(reserved)																UART_WK_CHAR0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	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UART_WK_CHAR0 Configures wakeup character 0. (R/W)

Register 42.14. UART_SLEEP_CONF2_REG (0x0038)

(reserved)				UART_WK_MODE_SEL				UART_WK_CHAR_MASK				UART_WK_CHAR_NUM				UART_RX_WAKE_UP_THRHD				UART_ACTIVE_THRESHOLD			
31	28	27	26	25	21	20	18	17	10	9	0												
0	0	0	0	0	0x0				0x5				1				0xf0				Reset		

UART_ACTIVE_THRESHOLD Configures the number of RXD edge changes to wake up the chip in wakeup mode 0. (R/W)

UART_RX_WAKE_UP_THRHD Configures the number of received data bytes to wake up the chip in wakeup mode 1. (R/W)

UART_WK_CHAR_NUM Configures the number of wakeup characters. (R/W)

UART_WK_CHAR_MASK Configures whether or not to mask wakeup characters.

0: Not mask

1: Mask

(R/W)

UART_WK_MODE_SEL Configures which wakeup mode to select. See Section [42.4.8 Wakeup](#) for the explanation of each mode.

0: Mode 0

1: Mode 1

2: Mode 2

3: Mode 3

(R/W)

Register 42.15. UART_SWFC_CONFO_SYNC_REG (0x003C)

(reserved)																UART_SEND_XOFF				UART_SEND_XON				UART_FORCE_XOFF				UART_FORCE_XON				UART_XONOFF_DEL				UART_SW_FLOW_CON_EN				UART_XON_XOFF_STILL_SEND				UART_XOFF_CHAR				UART_XON_CHAR			
31								23	22	21	20	19	18	17	16	15								8	7								0																		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x13							0x11							Reset																					

UART_XON_CHAR Configures the XON character for flow control. (R/W)

UART_XOFF_CHAR Configures the XOFF character for flow control. (R/W)

UART_XON_XOFF_STILL_SEND Configures whether the UART transmitter can send XON or XOFF characters when it is disabled.

0: Cannot send

1: Can send

(R/W)

UART_SW_FLOW_CON_EN Configures whether or not to enable software flow control.

0: Disable

1: Enable

(R/W)

UART_XONOFF_DEL Configures whether or not to remove flow control characters from the received data.

0: Not move

1: Move

(R/W)

UART_FORCE_XON Configures whether the transmitter continues to sending data.

0: Not send

1: Send

(R/W)

UART_FORCE_XOFF Configures whether or not to stop the transmitter from sending data.

0: Not stop

1: Stop

(R/W)

UART_SEND_XON Configures whether or not to send XON characters.

0: Not send

1: Send

(R/W/SS/SC)

UART_SEND_XOFF Configures whether or not to send XOFF characters.

0: Not send

1: Send

(R/W/SS/SC)

Register 42.16. UART_SWFC_CONF1_REG (0x0040)

(reserved)																UART_XOFF_THRESHOLD								UART_XON_THRESHOLD																															
31																16								15								8								7								0							
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0xe0								0x0								Reset																							

UART_XON_THRESHOLD Configures the threshold for data in RX FIFO to send XON characters in software flow control.

Measurement unit: byte. (R/W)

UART_XOFF_THRESHOLD Configures the threshold for data in RX FIFO to send XOFF characters in software flow control.

Measurement unit: byte. (R/W)

Register 42.17. UART_TXBRK_CONF_SYNC_REG (0x0044)

(reserved)																								UART_TX_BRK_NUM									
31																								8	7	0							
0 0																								Oxa								Reset	

UART_TX_BRK_NUM Configures the number of NULL characters to be sent after finishing data transmission.

Valid only when UART_TXD_BRK is 1. (R/W)

Register 42.18. UART_IDLE_CONF_SYNC_REG (0x0048)

(reserved)												UART_TX_IDLE_NUM								UART_RX_IDLE_THRHD																															
31												20								19								10								9								0							
0 0 0 0 0 0 0 0 0 0 0 0												0x100								0x100								Reset																							

UART_RX_IDLE_THRHD Configures the threshold to generate a frame end signal when the receiver takes more time to receive one data byte data.

Measurement unit: bit time (the time to transmit 1 bit). (R/W)

UART_TX_IDLE_NUM Configures the interval between two data transfers.

Measurement unit: bit time (the time to transmit 1 bit). (R/W)

Register 42.19. UART_RS485_CONF_SYNC_REG (0x004C)

[illegible]

UART_RS485_EN Configures whether or not to enable RS485 mode.

0: Disable

1: Enable

(R/W)

UART_DLO_EN Configures whether or not to add a turnaround delay of 1 bit before the start bit.

0: Not add

1: Add

(R/W)

UART_DL1_EN Configures whether or not to add a turnaround delay of 1 bit after the stop bit.

0: Not add

1: Add

(R/W)

UART_RS485TX_RX_EN Configures whether or not to enable the receiver for data reception when the transmitter is transmitting data in RS485 mode.

0: Disable

1: Enable

(R/W)

UART_RS485RXBY_TX_EN Configures whether to enable the RS485 transmitter for data transmission when the RS485 receiver is busy.

0: Disable

1: Enable

(R/W)

UART_RS485_RX_DLY_NUM Configures the delay of internal data signals in the receiver.

Measurement unit: bit time (the time to transmit 1 bit).. (R/W)

UART_RS485_TX_DLY_NUM Configures the delay of internal data signals in the transmitter.

Measurement unit: bit time (the time to transmit 1 bit). (R/W)

Register 42.20. UART_CLK_CONF_REG (0x0088)

[illegible]

UART_TX_SCLK_EN Configures whether or not to enable UART TX clock.

0: Disable

1: Enable

(R/W)

UART_RX_SCLK_EN Configures whether or not to enable UART RX clock.

0: Disable

1: Enable

(R/W)

UART_TX_RST_CORE Write 1 and then write 0 to reset UART TX. (R/W)

UART_RX_RST_CORE Write 1 and then write 0 to reset UART RX. (R/W)

Register 42.21. UART_STATUS_REG (0x001C)

The register map shows several UART-related registers:

- UART_TXD**, **UART_RTSN**, **UART_DTRN**: Located between 0x8C and 0x9F.
- (reserved)
- UART_TXIFIFO_CNT**: Located around 0xB6-0xBF.
- UART_RXD**, **UART_CTSN**, **UART_DSRRN**: Located between 0xC0 and 0xD3.
- (reserved)
- UART_RXIFIFO_CNT**: Located around 0xE6-0xEF.

31	30	29	28		24	23		16	15	14	13	12		8	7		0
1	1	1	0	0	0	0	0	0		1	1	0	0	0	0	0	0

Reset

UART_RXFIFO_CNT Represents the number of valid data bytes in RX FIFO. (RO)

UART_DSRN Represents the level of the internal UART DSR signal. (RO)

UART_CTSN Represents the level of the internal UART CTS signal. (RO)

UART_RXD Represents the level of the internal UART RXD signal. (RO)

UART_TXFIFO_CNT Represents the number of valid data bytes in RX FIFO. (RO)

UART_DTRN Represents the level of the internal UART DTR signal. (RO)

UART_RTSN Represents the level of the internal UART RTS signal. (RO)

UART_TXD Represents the level of the internal UART TXD signal. (RO)

Register 42.22. UART_MEM_TX_STATUS_REG (0x0068)

(reserved)																UART_TX_SRAM_RADDR								(reserved)								UART_TX_SRAM_WADDR																																							
31																17																16								9								8								7								0							
0																0																0x0								0								0x0								Reset															

UART_TX_SRAM_WADDR Represents the offset address to write TX FIFO. (RO)

UART_TX_SRAM_RADDR Represents the offset address to read TX FIFO. (RO)

Register 42.23. UART_MEM_RX_STATUS_REG (0x006C)

(reserved)																UART_RX_SRAM_WADDR								(reserved)								UART_RX_SRAM_RADDR																																																																																																	
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UART_RX_SRAM_RADDR Represents the offset address to read RX FIFO. (RO)

UART_RX_SRAM_WADDR Represents the offset address to write RX FIFO. (RO)

Register 42.24. UART_FSM_STATUS_REG (0x0070)

(reserved)																								UART_ST_UTX_OUT				UART_ST_URX_OUT																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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UART_ST_URX_OUT Represents the status of the receiver. (RO)

UART_ST_UTX_OUT Represents the status of the transmitter. (RO)

Register 42.25. UART_AFIFO_STATUS_REG (0x0090)

(reserved)																												UART_RX_AFIFO_EMPTY UART_RX_AFIFO_FULL UART_TX_AFIFO_EMPTY UART_TX_AFIFO_FULL				
31																												4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	Reset	

UART_TX_AFIFO_FULL Represents whether or not the APB TX asynchronous FIFO is full.

0: Not full

1: Full

(RO)

UART_TX_AFIFO_EMPTY Represents whether or not the APB TX asynchronous FIFO is empty.

0: Not empty

1: Empty

(RO)

UART_RX_AFIFO_FULL Represents whether or not the APB RX asynchronous FIFO is full.

0: Not full

1: Full

(RO)

UART_RX_AFIFO_EMPTY Represents whether or not the APB RX asynchronous FIFO is empty.

0: Not empty

1: Empty

(RO)

Register 42.26. UART_AT_CMD_PRECNT_SYNC_REG (0x0050)

(reserved)																UART_PRE_IDLE_NUM																															
31																15																0															
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0x901																Reset															

UART_PRE_IDLE_NUM Configures the idle time before the receiver receives the first AT_CMD.

Measurement unit: bit time (the time to transmit 1 bit). (R/W)

Register 42.27. UART_AT_CMD_POSTCNT_SYNC_REG (0x0054)

(reserved)																UART_POST_IDLE_NUM															
31																0															
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0x901															
Reset																															

UART_POST_IDLE_NUM Configures the interval between the last AT_CMD and subsequent data.
Measurement unit: bit time (the time to transmit 1 bit). (R/W)

Register 42.28. UART_AT_CMD_GAPTOUT_SYNC_REG (0x0058)

Register 0x00000000 (UART_RX_GAP_TOUT) bit field diagram:

- Bits 31-16: (reserved)
- Bits 15-0: UART_RX_GAP_TOUT (16 bits)
- Sub-fields:
 - UART_RX_GAP_TOUT[8:0] (8 bits)
 - UART_RX_GAP_TOUT[15:8] (8 bits)

UART_RX_GAP_TOUT Configures the interval between two AT_CMD characters.
Measurement unit: bit time (the time to transmit 1 bit). (R/W)

Register 42.29. UART_AT_CMD_CHAR_SYNC_REG (0x005C)

(reserved)																UART_CHAR_NUM								UART_AT_CMD_CHAR															
31																15								7								0							
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0x3								0x2b								Reset							

UART_AT_CMD_CHAR Configures the AT_CMD character. (R/W)

UART_CHAR_NUM Configures the number of continuous AT_CMD characters a receiver can receive.
(R/W)

Register 42.30. UART_POSPULSE_REG (0x0074)

Diagram illustrating the structure of the UART_PCR register. The register is 32 bits wide, divided into two 16-bit halves. The upper 16 bits (bits 31-16) are labeled "(reserved)". The lower 16 bits (bits 15-0) are labeled "UART_POSEDGE__MIN_CNT". The register is also labeled "Reset" on the right side.

UART_POSEDGE_MIN_CNT Represents the minimal input clock counter value between two positive edges. It is used for baud rate detection. (RO)

Register 42.31. UART_NEGPULSE_REG (0x0078)

(reserved)																UART_NEGEDGE_MIN_CNT															
31																0															
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0fff															
																Reset															

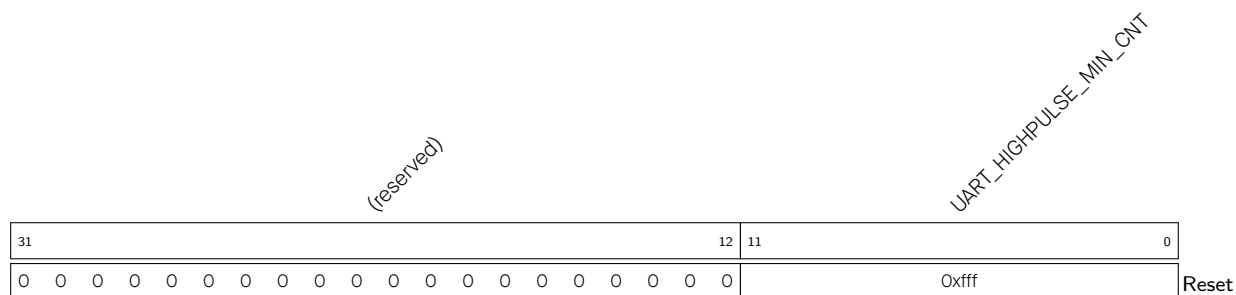
UART_NEGEDGE_MIN_CNT Represents the minimal input clock counter value between two negative edges. It is used for baud rate detection. (RO)

Register 42.32. UART_LOWPULSE_REG (0x007C)

31	12	11	0
(reserved)		UART_LOWPUSE_MIN_CNT	Reset
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0		0xff	

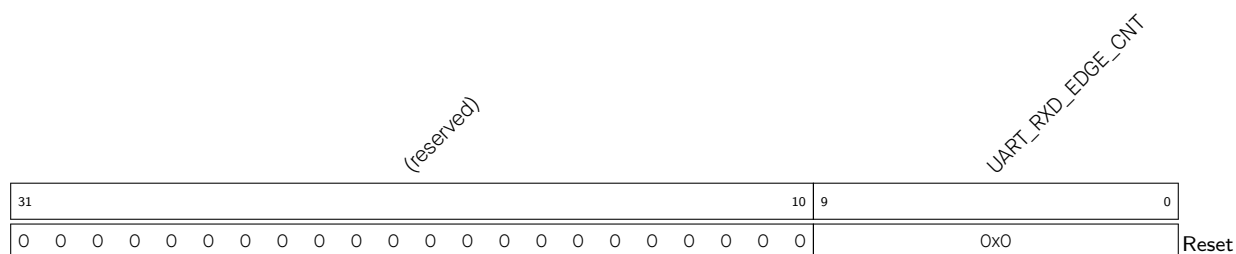
UART_LOWPULSE_MIN_CNT Represents the minimum duration time of a low-level pulse. It is used for baud rate detection.

Measurement unit: APB_CLK clock cycle. (RO)

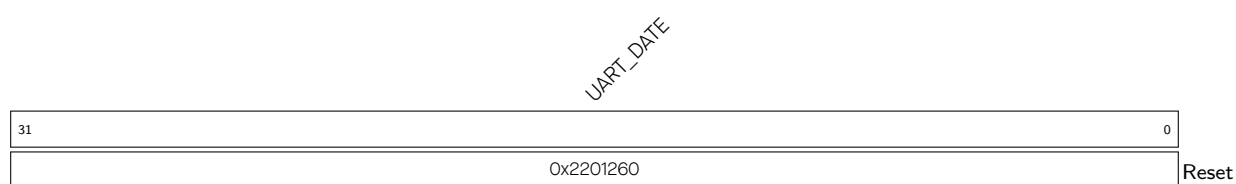
Register 42.33. UART_HIGHPULSE_REG (0x0080)

UART_HIGHPULSE_MIN_CNT Represents the maximum duration time for a high-level pulse. It is used for baud rate detection.

Measurement unit: APB_CLK clock cycle. (RO)

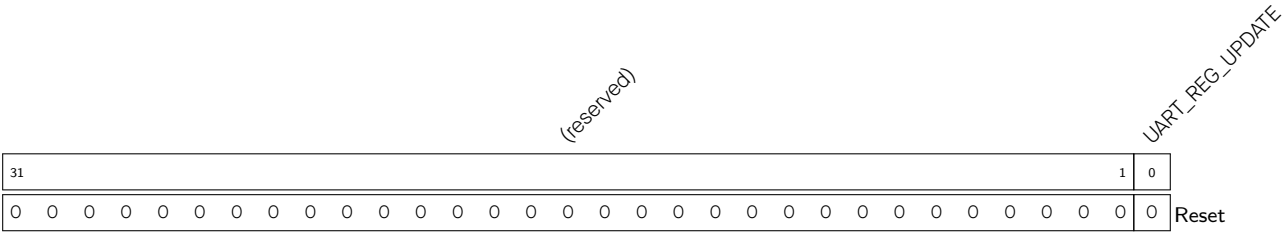
Register 42.34. UART_RXD_CNT_REG (0x0084)

UART_RXD_EDGE_CNT Represents the number of RXD edge changes. It is used for baud rate detection. (RO)

Register 42.35. UART_DATE_REG (0x008C)

UART_DATE Version control register. (R/W)

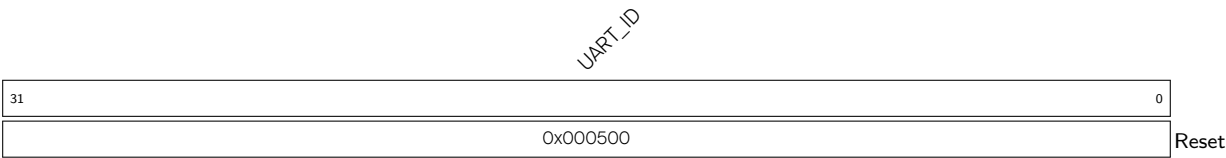
Register 42.36. UART_REG_UPDATE_REG (0x0098)



UART_REG_UPDATE Configures whether or not to synchronize registers.

- 0: Not synchronize
- 1: Synchronize (R/W/SC)

Register 42.37. UART_ID_REG (0x009C)



UART_ID Configures the UART ID. (R/W)

42.7.2 LP UART Registers

The addresses in this section are relative to LP UART base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

Register 42.38. LP_UART_FIFO_REG (0x0000)

31		7	0
(reserved)		LP_UART_RXFIFO_RD_BYTE	
0 0		0	
Reset			

LP_UART_RXFIFO_RD_BYTE Represents the data LP UART *n* read from FIFO.

Measurement unit: byte. (RO)

Register 42.39. LP_UART_TOUT_CONF_SYNC_REG (0x0064)

31	12	11	2	1	0
(reserved)		LP_UART_RX_TOUT_THRHD		LP_UART_RX_TOUT_FLOW_DIS	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0		0x0		0	0
				Reset	

LP_UART_RX_TOUT_EN Configures whether or not to enable LP UART receiver's timeout function.

0: Disable

1: Enable

(R/W)

LP_UART_RX_TOUT_FLOW_DIS Configures whether or not to disable the idle status counter when hardware flow control is enabled.

0: Invalid. No effect

1: Disable

(R/W)

LP_UART_RX_TOUT_THRHD Configures the amount of time that the bus can remain idle before time-out.

Measurement unit: bit time (the time to transmit 1 bit). (R/W)

Register 42.40. LP_UART_INT_RAW_REG (0x0004)

(reserved)																LP_UART_WAKEUP_INT_RAW		LP_UART_AT_CMD_CHAR_DET_INT_RAW		(reserved)																LP_UART_TX_DONE_INT_RAW		LP_UART_TX_BRK_IDLE_DONE_INT_RAW		LP_UART_TX_BRK_DONE_INT_RAW		LP_UART_GLITCH_DET_INT_RAW		LP_UART_SW_XOFF_INT_RAW		LP_UART_SW_XON_INT_RAW		LP_UART_RXFIFO_TOUT_INT_RAW		LP_UART_BRK_DET_INT_RAW		LP_UART_CTS_CHG_INT_RAW		LP_UART_DSR_CHG_INT_RAW		LP_UART_RXFIFO_OVF_INT_RAW		LP_UART_FRM_ERR_INT_RAW		LP_UART_PARTY_ERR_INT_RAW		LP_UART_TXFIFO_EMPTY_INT_RAW		LP_UART_RXFIFO_FULL_INT_RAW	
31													20	19	18	17	15		14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																																
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	Reset																																					

LP_UART_RXFIFO_FULL_INT_RAW The raw interrupt status of LP_UART_RXFIFO_FULL_INT. (R/WTC/SS)

LP_UART_TXFIFO_EMPTY_INT_RAW The raw interrupt status of LP_UART_TXFIFO_EMPTY_INT. (R/WTC/SS)

LP_UART_PARITY_ERR_INT_RAW The raw interrupt status of LP_UART_PARITY_ERR_INT. (R/WTC/SS)

LP_UART_FRM_ERR_INT_RAW The raw interrupt status of LP_UART_FRM_ERR_INT. (R/WTC/SS)

LP_UART_RXFIFO_OVF_INT_RAW The raw interrupt status of LP_UART_RXFIFO_OVF_INT. (R/WTC/SS)

LP_UART_DSR_CHG_INT_RAW The raw interrupt status of LP_UART_DSR_CHG_INT. (R/WTC/SS)

LP_UART_CTS_CHG_INT_RAW The raw interrupt status of LP_UART_CTS_CHG_INT. (R/WTC/SS)

LP_UART_BRK_DET_INT_RAW The raw interrupt status of LP_UART_BRK_DET_INT. (R/WTC/SS)

LP_UART_RXFIFO_TOUT_INT_RAW The raw interrupt status of LP_UART_RXFIFO_TOUT_INT. (R/WTC/SS)

LP_UART_SW_XON_INT_RAW The raw interrupt status of LP_UART_SW_XON_INT. (R/WTC/SS)

LP_UART_SW_XOFF_INT_RAW LP_UART_SW_XOFF_INT. (R/WTC/SS)

LP_UART_GLITCH_DET_INT_RAW The raw interrupt status of LP_UART_GLITCH_DET_INT. (R/WTC/SS)

LP_UART_TX_BRK_DONE_INT_RAW The raw interrupt status of LP_UART_TX_BRK_DONE_INT. (R/WTC/SS)

LP_UART_TX_BRK_IDLE_DONE_INT_RAW The raw interrupt status of LP_UART_TX_BRK_IDLE_DONE_INT. (R/WTC/SS)

Continued on the next page...

Register 42.40. LP_UART_INT_RAW_REG (0x0004)

Continued from the previous page...

LP_UART_TX_DONE_INT_RAW The raw interrupt status of LP_UART_TX_DONE_INT. (R/WTC/SS)

LP_UART_AT_CMD_CHAR_DET_INT_RAW The raw interrupt status of LP_UART_AT_CMD_CHAR_DET_INT. (R/WTC/SS)

LP_UART_WAKEUP_INT_RAW The raw interrupt status of LP_UART_WAKEUP_INT. (R/WTC/SS)

Register 42.41. LP_UART_INT_ST_REG (0x0008)

(reserved)																LP_UART_WAKEUP_INT_ST LP_UART_AT_CMD_CHAR_DET_INT_ST (reserved)																LP_UART_TX_DONE_INT_ST LP_UART_TX_BRK_IDLE_DONE_INT_ST LP_UART_TX_BRK_DONE_INT_ST LP_UART_GLITCH_DET_INT_ST LP_UART_SW_XON_INT_ST LP_UART_SW_XOFF_INT_ST LP_UART_RXFIFO_TOUT_INT_ST LP_UART_CTS_CHG_INT_ST LP_UART_DSR_CHG_INT_ST LP_UART_RXFIFO_ERR_INT_ST LP_UART_FRM_ERR_INT_ST LP_UART_PARITY_ERR_INT_ST LP_UART_RXFIFO_EMPTY_INT_ST LP_UART_RXFIFO_FULL_INT_ST															
31													20	19	18	17	15		14	13	12	11	10	9	8	7	6	5	4	3	2	1	0														
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset																			

LP_UART_RXFIFO_FULL_INT_ST The masked interrupt status of LP_UART_RXFIFO_FULL_INT. (RO)

LP_UART_TXFIFO_EMPTY_INT_ST The masked interrupt status of LP_UART_TXFIFO_EMPTY_INT. (RO)

LP_UART_PARITY_ERR_INT_ST The masked interrupt status of LP_UART_PARITY_ERR_INT. (RO)

LP_UART_FRM_ERR_INT_ST The masked interrupt status of LP_UART_FRM_ERR_INT. (RO)

LP_UART_RXFIFO_OVF_INT_ST The masked interrupt status of LP_UART_RXFIFO_OVF_INT. (RO)

LP_UART_DSR_CHG_INT_ST The masked interrupt status of LP_UART_DSR_CHG_INT. (RO)

LP_UART_CTS_CHG_INT_ST The masked interrupt status of LP_UART_CTS_CHG_INT. (RO)

LP_UART_BRK_DET_INT_ST The masked interrupt status of LP_UART_BRK_DET_INT. (RO)

LP_UART_RXFIFO_TOUT_INT_ST The masked interrupt status of LP_UART_RXFIFO_TOUT_INT. (RO)

LP_UART_SW_XON_INT_ST The masked interrupt status of LP_UART_SW_XON_INT. (RO)

LP_UART_SW_XOFF_INT_ST The masked interrupt status of LP_UART_SW_XOFF_INT. (RO)

LP_UART_GLITCH_DET_INT_ST The masked interrupt status of LP_UART_GLITCH_DET_INT. (RO)

LP_UART_TX_BRK_DONE_INT_ST The masked interrupt status of LP_UART_TX_BRK_DONE_INT. (RO)

LP_UART_TX_BRK_IDLE_DONE_INT_ST The masked interrupt status of LP_UART_TX_BRK_IDLE_DONE_INT. (RO)

LP_UART_TX_DONE_INT_ST The masked interrupt status of LP_UART_TX_DONE_INT. (RO)

LP_UART_AT_CMD_CHAR_DET_INT_ST The masked interrupt status of LP_UART_AT_CMD_CHAR_DET_INT. (RO)

LP_UART_WAKEUP_INT_ST The masked interrupt status of LP_UART_WAKEUP_INT. (RO)

Register 42.42. LP_UART_INT_ENA_REG (0x000C)

(reserved)												LP_UART_WAKEUP_INT_ENA LP_UART_AT_CMD_CHAR_DET_INT_ENA (reserved) LP_UART_TX_DONE_INT_ENA LP_UART_TX_BRK_IDLE_DONE_INT_ENA LP_UART_TX_BRK_DONE_INT_ENA LP_UART_GLITCH_DET_INT_ENA LP_UART_SW_XON_INT_ENA LP_UART_SW_XOFF_INT_ENA LP_UART_RXFIFO_TOUT_INT_ENA LP_UART_BRK_DET_INT_ENA LP_UART_CTS_CHG_INT_ENA LP_UART_DSR_CHG_INT_ENA LP_UART_RXFIFO_OVF_INT_ENA LP_UART_FRM_ERR_INT_ENA LP_UART_PARITY_ERR_INT_ENA LP_UART_TXFIFO_EMPTY_INT_ENA LP_UART_RXFIFO_FULL_INT_ENA																					
31													20	19	18	17	15		14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset				

LP_UART_RXFIFO_FULL_INT_ENA Write 1 to enable LP_UART_RXFIFO_FULL_INT. (R/W)

LP_UART_TXFIFO_EMPTY_INT_ENA Write 1 to enable LP_UART_TXFIFO_EMPTY_INT. (R/W)

LP_UART_PARITY_ERR_INT_ENA Write 1 to enable LP_UART_PARITY_ERR_INT. (R/W)

LP_UART_FRM_ERR_INT_ENA Write 1 to enable LP_UART_FRM_ERR_INT. (R/W)

LP_UART_RXFIFO_OVF_INT_ENA Write 1 to enable LP_UART_RXFIFO_OVF_INT. (R/W)

LP_UART_DSR_CHG_INT_ENA Write 1 to enable LP_UART_DSR_CHG_INT. (R/W)

LP_UART_CTS_CHG_INT_ENA Write 1 to enable LP_UART_CTS_CHG_INT. (R/W)

LP_UART_BRK_DET_INT_ENA Write 1 to enable LP_UART_BRK_DET_INT. (R/W)

LP_UART_RXFIFO_TOUT_INT_ENA Write 1 to enable LP_UART_RXFIFO_TOUT_INT. (R/W)

LP_UART_SW_XON_INT_ENA Write 1 to enable LP_UART_SW_XON_INT. (R/W)

LP_UART_SW_XOFF_INT_ENA Write 1 to enable LP_UART_SW_XOFF_INT. (R/W)

LP_UART_GLITCH_DET_INT_ENA Write 1 to enable LP_UART_GLITCH_DET_INT. (R/W)

LP_UART_TX_BRK_DONE_INT_ENA Write 1 to enable LP_UART_TX_BRK_DONE_INT. (R/W)

LP_UART_TX_BRK_IDLE_DONE_INT_ENA Write 1 to enable LP_UART_TX_BRK_IDLE_DONE_INT. (R/W)

LP_UART_TX_DONE_INT_ENA Write 1 to enable LP_UART_TX_DONE_INT. (R/W)

LP_UART_AT_CMD_CHAR_DET_INT_ENA Write 1 to enable LP_UART_AT_CMD_CHAR_DET_INT. (R/W)

LP_UART_WAKEUP_INT_ENA Write 1 to enable LP_UART_WAKEUP_INT. (R/W)

Register 42.43. LP_UART_INT_CLR_REG (0x0010)

(reserved)																LP_UART_WAKEUP_INT_CLR LP_UART_AT_CMD_CHAR_DET_INT_CLR (reserved) LP_UART_TX_DONE_INT_CLR LP_UART_TX_BRK_IDLE_DONE_INT_CLR LP_UART_TX_BRK_DONE_INT_CLR LP_UART_GLITCH_DET_INT_CLR LP_UART_SW_XOFF_INT_CLR LP_UART_SW_XON_INT_CLR LP_UART_RXFIFO_TOUT_INT_CLR LP_UART_BRK_DET_INT_CLR LP_UART_CTS_CHG_INT_CLR LP_UART_DSR_CHG_INT_CLR LP_UART_RXFIFO_OVF_INT_CLR LP_UART_FRM_ERR_INT_CLR LP_UART_PARITY_ERR_INT_CLR LP_UART_TXFIFO_EMPTY_INT_CLR LP_UART_RXFIFO_FULL_INT_CLR																		
31													20	19	18	17	15		14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						

LP_UART_RXFIFO_FULL_INT_CLR Write 1 to clear LP_UART_RXFIFO_FULL_INT. (WT)

LP_UART_TXFIFO_EMPTY_INT_CLR Write 1 to clear LP_UART_TXFIFO_EMPTY_INT. (WT)

LP_UART_PARITY_ERR_INT_CLR Write 1 to clear LP_UART_PARITY_ERR_INT. (WT)

LP_UART_FRM_ERR_INT_CLR Write 1 to clear LP_UART_FRM_ERR_INT. (WT)

LP_UART_RXFIFO_OVF_INT_CLR Write 1 to clear LP_UART_RXFIFO_OVF_INT. (WT)

LP_UART_DSR_CHG_INT_CLR Write 1 to clear LP_UART_DSR_CHG_INT. (WT)

LP_UART_CTS_CHG_INT_CLR Write 1 to clear LP_UART_CTS_CHG_INT. (WT)

LP_UART_BRK_DET_INT_CLR Write 1 to clear LP_UART_BRK_DET_INT. (WT)

LP_UART_RXFIFO_TOUT_INT_CLR Write 1 to clear LP_UART_RXFIFO_TOUT_INT. (WT)

LP_UART_SW_XON_INT_CLR Write 1 to clear LP_UART_SW_XON_INT. (WT)

LP_UART_SW_XOFF_INT_CLR Write 1 to clear LP_UART_SW_XOFF_INT. (WT)

LP_UART_GLITCH_DET_INT_CLR Write 1 to clear LP_UART_GLITCH_DET_INT. (WT)

LP_UART_TX_BRK_DONE_INT_CLR Write 1 to clear LP_UART_TX_BRK_DONE_INT. (WT)

LP_UART_TX_BRK_IDLE_DONE_INT_CLR Write 1 to clear LP_UART_TX_BRK_IDLE_DONE_INT. (WT)

LP_UART_TX_DONE_INT_CLR Write 1 to clear LP_UART_TX_DONE_INT. (WT)

LP_UART_AT_CMD_CHAR_DET_INT_CLR Write 1 to clear LP_UART_AT_CMD_CHAR_DET_INT. (WT)

LP_UART_WAKEUP_INT_CLR Write 1 to clear LP_UART_WAKEUP_INT. (WT)

Register 42.44. LP_UART_CLKDIV_SYNC_REG (0x0014)

(reserved)								LP_UART_CLKDIV_FRAG								(reserved)								LP_UART_CLKDIV																																																																																															
31								24								23								20								19								12								11								0																																																															
0								0								0								0								0								0								0								0x0								0								0								0								0								0								0x2b6								Reset							

LP_UART_CLKDIV Configures the integral part of the divisor for baud rate generation. (R/W)

LP_UART_CLKDIV_FRAG Configures the fractional part of the divisor for baud rate generation. (R/W)

Register 42.45. LP_UART_RX_FILT_REG (0x0018)

(reserved)																								LP_UART_GLITCH_FILT_EN		LP_UART_GLITCH_FILT		
31																								9	8	7	0	
0 0																								0	0x8		Reset	

LP_UART_GLITCH_FILT Configures the width of a pulse to be filtered.

Measurement unit: UART Core's clock cycle.

Pulses whose width is lower than this value will be ignored. (R/W)

LP_UART_GLITCH_FILT_EN Configures whether or not to enable RX signal filter.

0: Disable

1: Enable

(R/W)

Register 42.46. LP_UART_CONFO_SYNC_REG (0x0020)

(reserved)								LP_UART_TXFIFO_RST LP_UART_RXFIFO_RST LP_UART_SW_RST LP_UART_MEM_CLK_EN (reserved) LP_UART_ERR_WR_MASK LP_UART_DIS_RX_DAT_OVF LP_UART_TXD_INV (reserved) LP_UART_RXD_INV LP_UART_TX_FLOW_EN LP_UART_LOOPBACK											(reserved) LP_UART_TXD_BRK LP_UART_STOP_BIT_NUM LP_UART_BIT_NUM LP_UART_PARITY_EN LP_UART_PARITY												
31								24	23	22	21	20	19	18	17	16	15	14	13	12	11			7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	3	0	0	Reset	

LP_UART_PARITY Configures the parity check mode.

0: Even parity

1: Odd parity

(R/W)

LP_UART_PARITY_EN Configures whether or not to enable LP UART parity check.

0: Disable

1: Enable

(R/W)

LP_UART_BIT_NUM Configures the number of data bits.

0: 5 bits

1: 6 bits

2: 7 bits

3: 8 bits

(R/W)

LP_UART_STOP_BIT_NUM Configures the number of stop bits.

0: Invalid. No effect

1: 1 bit

2: 1.5 bits

3: 2 bits

(R/W)

LP_UART_TXD_BRK Configures whether or not to send NULL characters when finishing data transmission.

0: Not send

1: Send

(R/W)

LP_UART_LOOPBACK Configures whether or not to enable LP UART loopback test.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 42.46. LP_UART_CONFO_SYNC_REG (0x0020)

Continued from the previous page...

LP_UART_TX_FLOW_EN Configures whether or not to enable flow control for the transmitter.

0: Disable

1: Enable

(R/W)

LP_UART_RXD_INV Configures whether or not to invert the level of LP UART RXD signal.

0: Not invert

1: Invert

(R/W)

LP_UART_TXD_INV Configures whether or not to invert the level of LP UART TXD signal.

0: Not invert

1: Invert

(R/W)

LP_UART_DIS_RX_DAT_OVF Configures whether or not to disable data overflow detection for the LP UART receiver.

0: Enable

1: Disable

(R/W)

LP_UART_ERR_WR_MASK Configures whether or not to store the received data with errors into FIFO.

0: Store

1: Not store

(R/W)

LP_UART_MEM_CLK_EN Configures whether or not to enable clock gating for LP UART memory.

0: Disable

1: Enable

(R/W)

LP_UART_SW_RTS Configures the RTS signal used in software flow control.

0: The LP UART transmitter is not allowed to send data.

1: The LP UART transmitted is allowed to send data.

(R/W)

LP_UART_RXFIFO_RST Configures whether or not to reset the LP UART RX FIFO.

0: Not reset

1: Reset

(R/W)

LP_UART_TXFIFO_RST Configures whether or not to reset the LP UART TX FIFO.

0: Not reset

1: Reset

(R/W)

Register 42.47. LP_UART_CONF1_REG (0x0024)

(reserved)												LP_UART_CLK_EN				LP_UART_SW_DTR				LP_UART_DTR_INV				LP_UART_RTS_INV				LP_UART_DSR_INV				LP_UART_CTS_INV				LP_UART_TXFIFO_EMPTY_THRHD				(reserved)				LP_UART_RXFIFO_FULL_THRHD				(reserved)			
31												22	21	20	19	18	17	16	15	11				10	8		7	3				2	0																		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0xc	0xc				0	0	0	0xc	0xc				0	0	0	Reset																			

LP_UART_RXFIFO_FULL_THRHD Configures the threshold for RX FIFO being full.

Measurement unit: byte. (R/W)

LP_UART_TXFIFO_EMPTY_THRHD Configures the threshold for TX FIFO being empty.

Measurement unit: byte. (R/W)

LP_UART_CTS_INV Configures whether or not to invert the level of LP UART CTS signal.

0: Not invert

1: Invert

(R/W)

LP_UART_DSR_INV Configures whether or not to invert the level of LP UART DSR signal.

0: Not invert

1: Invert

(R/W)

LP_UART_RTS_INV Configures whether or not to invert the level of LP UART RTS signal.

0: Not invert

1: Invert

(R/W)

LP_UART_DTR_INV Configures whether or not to invert the level of LP UART DTR signal.

0: Not invert

1: Invert

(R/W)

LP_UART_SW_DTR Configures the DTR signal used in software flow control.

0: Data to be transmitted is not ready.

1: Data to be transmitted is ready.

(R/W)

LP_UART_CLK_EN Configures clock gating.

0: Support clock only when the application writes registers.

1: Always force the clock on for registers.

(R/W)

Register 42.48. LP_UART_HWFC_CONF_SYNC_REG (0x002C)

(reserved)																								LP_UART_RX_FLOW_EN		LP_UART_RX_FLOW_THRHD		(reserved)				
31																								9	8	7	3			2	0	
0 0																								0	0x0			0 0 0		Reset		

LP_UART_RX_FLOW_THRHD Configures the maximum number of data bytes that can be received during hardware flow control.

Measurement unit: byte. (R/W)

LP_UART_RX_FLOW_EN Configures whether or not to enable the LP UART receiver.

0: Disable

1: Enable

(R/W)

Register 42.49. LP_UART_SLEEP_CONFO_REG (0x0030)

LP_UART_WK_CHAR4								LP_UART_WK_CHAR3								LP_UART_WK_CHAR2								LP_UART_WK_CHAR1								
31					24	23					16	15					8	7					0									
0x0								0x0								0x0								0x0								Reset

LP_UART_WK_CHAR1 Configures wakeup character 1. (R/W)

LP_UART_WK_CHAR2 Configures wakeup character 2. (R/W)

LP_UART_WK_CHAR3 Configures wakeup character 3. (R/W)

LP_UART_WK_CHAR4 Configures wakeup character 4. (R/W)

Register 42.50. LP_UART_SLEEP_CONF1_REG (0x0034)

(reserved)																LP_UART_WK_CHAR0																	
31																8	7	0															
0 0																0x0																Reset	

LP_UART_WK_CHAR0 Configures wakeup character 0. (R/W)

Register 42.51. LP_UART_SLEEP_CONF2_REG (0x0038)

(reserved)				LP_UART_WK_MODE_SEL				LP_UART_WK_CHAR_MASK				LP_UART_WK_CHAR_NUM				LP_UART_RX_WAKE_UP_THRHD				(reserved)				LP_UART_ACTIVE_THRESHOLD			
31		28		27	26	25		21		20	18		17	13		12	10		9	0							
0	0	0	0	0	0	0x0				0x5				1				0	0	0	0xf0				Reset		

LP_UART_ACTIVE_THRESHOLD Configures the number of RXD edge changes to wake up the chip in wakeup mode 0. (R/W)

LP_UART_RX_WAKE_UP_THRHD Configures the number of received data bytes to wake up the chip in wakeup mode 1. (R/W)

LP_UART_WK_CHAR_NUM Configures the number of wakeup characters. (R/W)

LP_UART_WK_CHAR_MASK Configures whether or not to mask wakeup characters.

0: Not mask

1: Mask

(R/W)

LP_UART_WK_MODE_SEL Configures which wakeup mode to select.

0: Mode 0

1: Mode 1

2: Mode 2

3: Mode 3

(R/W)

Register 42.52. LP_UART_SWFC_CONFO_SYNC_REG (0x003C)

(reserved)										LP_UART_SEND_XOFF								LP_UART_SEND_XON								LP_UART_FORCE_XOFF								LP_UART_FORCE_XON								LP_UART_SW_FLOW_CON_EN								LP_UART_XON_XOFF_STILL_SEND								LP_UART_XOFF_CHAR								LP_UART_XON_CHAR							
31								23	22	21	20	19	18	17	16	15								8	7								0																																								
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x13							0x11							Reset																																											

LP_UART_XON_CHAR Configures the XON character for flow control. (R/W)

LP_UART_XOFF_CHAR Configures the XOFF character for flow control. (R/W)

LP_UART_XON_XOFF_STILL_SEND Configures whether the LP UART transmitter can send XON or XOFF characters when it is disabled.

0: Cannot send

1: Can send

(R/W)

LP_UART_SW_FLOW_CON_EN Configures whether or not to enable software flow control.

0: Disable

1: Enable

(R/W)

LP_UART_XONOFF_DEL Configures whether or not to remove flow control characters from the received data.

0: Not move

1: Move

(R/W)

LP_UART_FORCE_XON Configures whether the transmitter continues to sending data.

0: Not send

1: Send

(R/W)

LP_UART_FORCE_XOFF Configures whether or not to stop the transmitter from sending data.

0: Not stop

1: Stop

(R/W)

LP_UART_SEND_XON Configures whether or not to send XON characters.

0: Not send

1: Send

(R/W/SS/SC)

LP_UART_SEND_XOFF Configures whether or not to send XOFF characters.

0: Not send

1: Send

(R/W/SS/SC)

Register 42.53. LP_UART_SWFC_CONF1_REG (0x0040)

(reserved)																LP_UART_XOFF_THRESHOLD								(reserved)								LP_UART_XON_THRESHOLD								(reserved)																																																																																																																																																																																																																																																							
31																16																15																11																10																8																7																3																2																0																																																																																																																																															
0																0																0																0																0																0																0																0																0																0xc																0																0																0																0x0																0																0																0																Reset															

LP_UART_XON_THRESHOLD Configures the threshold for data in RX FIFO to send XON characters in software flow control.

Measurement unit: byte. (R/W)

LP_UART_XOFF_THRESHOLD Configures the threshold for data in RX FIFO to send XOFF characters in software flow control.

Measurement unit: byte. (R/W)

Register 42.54. LP_UART_TXBRK_CONF_SYNC_REG (0x0044)

(reserved)																								LP_UART_TX_BRK_NUM									
31																								8	7	0							
0 0																								0xa								Reset	

LP_UART_TX_BRK_NUM Configures the number of NULL characters to be sent after finishing data transmission.

Valid only when LP_UART_TXD_BRK is 1. (R/W)

Register 42.55. LP_UART_IDLE_CONF_SYNC_REG (0x0048)

(reserved)												LP_UART_TX_IDLE_NUM												LP_UART_RX_IDLE_THRHD																					
31											20											19											10	9											0
0	0	0	0	0	0	0	0	0	0	0	0	0x100												0x100												Reset									

LP_UART_RX_IDLE_THRHD Configures the threshold to generate a frame end signal when the receiver takes more time to receive one data byte data.

Measurement unit: bit time (the time to transmit 1 bit). (R/W)

LP_UART_TX_IDLE_NUM Configures the interval between two data transfers.

Measurement unit: bit time (the time to transmit 1 bit). (R/W)

Register 42.56. LP_UART_CLK_CONF_REG (0x0088)

(reserved)																LP_UART_RX_RST_CORE																LP_UART_TX_RST_CORE																LP_UART_RX_SCLK_EN																LP_UART_TX_SCLK_EN																(reserved)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
31				28				27		26		25		24		23																										0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
0		0		0		0		0		0		1		1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									

LP_UART_TX_SCLK_EN Configures whether or not to enable LP UART TX clock.

0: Disable

1: Enable

(R/W)

LP_UART_RX_SCLK_EN Configures whether or not to enable LP UART RX clock.

0: Disable

1: Enable

(R/W)

LP_UART_TX_RST_CORE Write 1 and then write 0 to reset LP UART TX. (R/W)

LP_UART_RX_RST_CORE Write 1 and then write 0 to reset LP UART RX. (R/W)

Register 42.57. LP_UART_STATUS_REG (0x001C)

LP_UART_TXD LP_UART_RTSN LP_UART_DTRN (reserved)				LP_UART_TXFIFO_CNT (reserved)				LP_UART_RXD LP_UART_CTSN LP_UART_DSRN (reserved)				LP_UART_RXFIFO_CNT (reserved)					
31	30	29	28	24	23	19	18	16	15	14	13	12	8	7	3	2	0
1	1	1	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0

Reset

Reset

LP_UART_RXFIFO_CNT Represents the number of valid data bytes in RX FIFO. (RO)**LP_UART_DSRN** Represents the level of the internal LP UART DSR signal. (RO)**LP_UART_CTSN** Represents the level of the internal LP UART CTS signal. (RO)**LP_UART_RXD** Represents the level of the internal LP UART RXD signal. (RO)**LP_UART_TXFIFO_CNT** Represents the number of valid data bytes in RX FIFO. (RO)**LP_UART_DTRN** Represents the level of the internal LP UART DTR signal. (RO)**LP_UART_RTSN** Represents the level of the internal LP UART RTS signal. (RO)**LP_UART_TXD** Represents the level of the internal LP UART TXD signal. (RO)**Register 42.58. LP_UART_MEM_TX_STATUS_REG (0x0068)**

(reserved)																LP_UART_TX_SRAM_RADDR				(reserved)				LP_UART_TX_SRAM_WADDR				(reserved)			
311716121187320																Reset															
0000000000000000																0x0				0000				0x0				0000			

Reset

LP_UART_TX_SRAM_WADDR Represents the offset address to write TX FIFO. (RO)**LP_UART_TX_SRAM_RADDR** Represents the offset address to read TX FIFO. (RO)

Register 42.59. LP_UART_MEM_RX_STATUS_REG (0x006C)

(reserved)																LP_UART_RX_SRAM_WADDR								(reserved)				LP_UART_RX_SRAM_RADDR				(reserved)			
311716121187320																																			
0000000000000000																0x10								0000				0x10				0000Reset			

Reset

LP_UART_RX_SRAM_RADDR Represents the offset address to read RX FIFO. (RO)

LP_UART_RX_SRAM_WADDR Represents the offset address to write RX FIFO. (RO)

Register 42.60. LP_UART_FSM_STATUS_REG (0x0070)

(reserved)																								LP_UART_ST_UTX_OUT				LP_UART_ST_URX_OUT			
31																								8	7	4		3	0		Reset
0 0																								0		0					

Reset

LP_UART_ST_URX_OUT Represents the status of the receiver. (RO)

LP_UART_ST_UTX_OUT Represents the status of the transmitter. (RO)

Register 42.61. LP_UART_AFIFO_STATUS_REG (0x0090)

(reserved)																												LP_UART_RX_AFIFO_EMPTY LP_UART_RX_AFIFO_FULL LP_UART_TX_AFIFO_EMPTY LP_UART_TX_AFIFO_FULL				
31																												4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	Reset	

Reset

LP_UART_TX_AFIFO_FULL Represents whether or not the APB TX asynchronous FIFO is full.

0: Not full

1: Full

(RO)

LP_UART_TX_AFIFO_EMPTY Represents whether or not the APB TX asynchronous FIFO is empty.

0: Not empty

1: Empty

(RO)

LP_UART_RX_AFIFO_FULL Represents whether or not the APB RX asynchronous FIFO is full.

0: Not full

1: Full

(RO)

LP_UART_RX_AFIFO_EMPTY Represents whether or not the APB RX asynchronous FIFO is empty.

0: Not empty

1: Empty

(RO)

Register 42.62. LP_UART_AT_CMD_PRECNT_SYNC_REG (0x0050)

(reserved)																LP_UART_PRE_IDLE_NUM																															
31																15																0															
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0x901																Reset															

Reset

LP_UART_PRE_IDLE_NUM Configures the idle time before the receiver receives the first AT_CMD.

Measurement unit: bit time (the time to transmit 1 bit). (R/W)

Register 42.63. LP_UART_AT_CMD_POSTCNT_SYNC_REG (0x0054)

(reserved)																LP_UART_POST_IDLE_NUM															
31																16	15														0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x901															
																Reset															

Reset

LP_UART_POST_IDLE_NUM Configures the interval between the last AT_CMD and subsequent data.
Measurement unit: bit time (the time to transmit 1 bit). (R/W)

Register 42.64. LP_UART_AT_CMD_GAPTOUT_SYNC_REG (0x0058)

(reserved)																LP_UART_RX_GAP_TOUT																															
31																15																0															
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																11																Reset															

Reset

LP_UART_RX_GAP_TOUT Configures the interval between two AT_CMD characters.
Measurement unit: bit time (the time to transmit 1 bit). (R/W)

Register 42.65. LP_UART_AT_CMD_CHAR_SYNC_REG (0x005C)

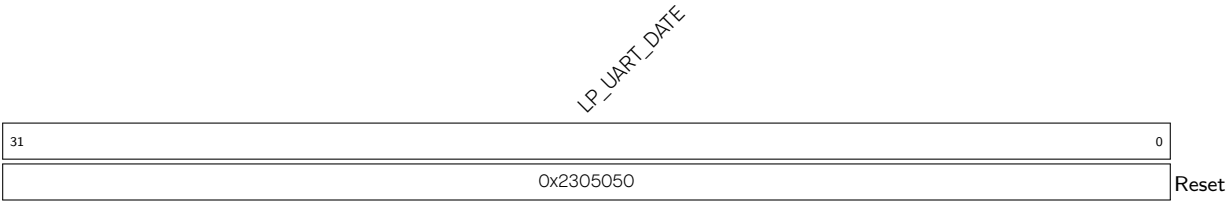
(reserved)																LP_UART_CHAR_NUM								LP_UART_AT_CMD_CHAR																																							
31																16																15								8								7								0							
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0x3																0x2b																Reset															

Reset

LP_UART_AT_CMD_CHAR Configures the AT_CMD character. (R/W)

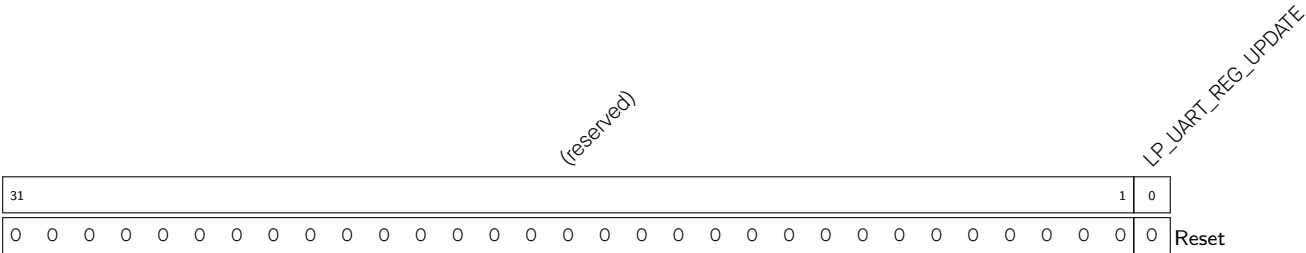
LP_UART_CHAR_NUM Configures the number of continuous AT_CMD characters a receiver can receive. (R/W)

Register 42.66. LP_UART_DATE_REG (0x008C)



LP_UART_DATE Version control register. (R/W)

Register 42.67. LP_UART_REG_UPDATE_REG (0x0098)

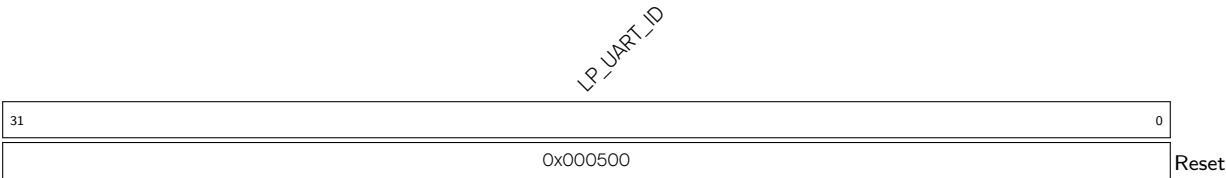


LP_UART_REG_UPDATE Configures whether or not to synchronize registers.

- 0: Not synchronize
- 1: Synchronize

(R/W/SC)

Register 42.68. LP_UART_ID_REG (0x009C)



LP_UART_ID Configures the LP UART ID. (R/W)

42.7.3 UHCI Registers

The addresses in this section are relative to UHCI base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

Register 42.69. UHCI_CONFO_REG (0x0000)

(reserved)																				UHCI_UART_RX_BRK_EOF_EN UHCI_CLK_EN UHCI_ENCODE_CRC_EN UHCI_LEN_EOF_EN UHCI_UART_IDLE_EOF_EN UHCI_CRC_REC_EN UHCI_HEAD_EN UHCI_SEPER_EN UHCI_UART_SEL UHCI_RX_RST UHCI_TX_RST									
31													13	12	11	10	9	8	7	6	5	4	2		1	0			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

Reset

UHCI_TX_RST Write 1 and then write 0 to reset the decoder state machine. (R/W)

UHCI_RX_RST Write 1 and then write 0 to reset the encoder state machine. (R/W)

UHCI_UART0_CE Select one UART from UART 0 ~ 4 to connect with UHCI.

0: Select UART0

1: Select UART1

2: Select UART2

3: Select UART3

4: Select UART4

5 ~ 7: No effect, as no UART will connect with UHCI

(R/W)

UHCI_UART1_CE Configures whether or not to connect UHCI with UART1.

0: Not connect

1: Connect

(R/W)

UHCI_SEPER_EN Configures whether or not to separate the data frame with a special character.

0: Not separate

1: Separate

(R/W)

UHCI_HEAD_EN Configures whether or not to encode the data packet with a formatting header.

0: Not use formatting header

1: Use formatting header

(R/W)

Continued on the next page...

Register 42.69. UHCI_CONFO_REG (0x0000)

Continued from the previous page...

UHCI_CRC_REC_EN Configures whether or not to enable the reception of the 16-bit CRC.

0: Disable

1: Enable

(R/W)

UHCI_UART_IDLE_EOF_EN Configures whether or not to stop receiving data when UART is idle.

0: Not stop

1: Stop

(R/W)

UHCI_LEN_EOF_EN Configures when the UHCI decoder stops receiving data.

0: Stops after receiving 0xC0

1: Stops when the number of received data bytes reach the specified value. When UHCI_HEAD_EN is 1, the specified value is the data length indicated by the UHCI packet header; when UHCI_HEAD_EN is 0, the specified value is the configured value.

(R/W)

UHCI_ENCODE_CRC_EN Configures whether or not to enable data integrity check by appending a 16 bit CCITT-CRC to the end of the data.

0: Disable

1: Enable

(R/W)

UHCI_CLK_EN Configures clock gating.

0: Support clock only when the application writes registers.

1: Always force the clock on for registers.

(R/W)

UHCI_UART_RX_BRK_EOF_EN Configures whether or not to stop UHCI from receiving data after UART has received a NULL frame.

0: Not stop

1: Stop

(R/W)

Register 42.70. UHCI_CONF1_REG (0x0014)

(reserved)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
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UHCI_CHECK_SUM_EN Configures whether or not to enable header checksum validation when UHCI receives a data packet.

0: Disable

1: Enable

(R/W)

UHCI_CHECK_SEQ_EN Configures whether or not to enable the sequence number check when UHCI receives a data packet.

0: Disable

1: Enable

(R/W)

UHCI_CRC_DISABLE Configures whether or not to enable CRC calculation.

0: Disable

1: Enable

Valid only when the Data Integrity Check Present bit in UHCI packet is 1.

(R/W)

UHCI_SAVE_HEAD Configures whether or not to save the packet header when UHCI receives a data packet.

0: Not save

1: Save

(R/W)

UHCI_TX_CHECK_SUM_RE Configures whether or not to encode the data packet with a checksum.

0: Not use checksum

1: Use checksum

(R/W)

UHCI_TX_ACK_NUM_RE Configures whether or not to encode the data packet with an acknowledgment when a reliable packet is to be transmitted.

0: Not use acknowledgement

1: Use acknowledgement

(R/W)

Continued on the next page...

Register 42.70. UHCI_CONF1_REG (0x0014)

Continued from the previous page...

UHCI_WAIT_SW_START Configures whether or not to put the UHCI encoder state machine to ST_SW_WAIT state.

0: No

1: Yes

(R/W)

UHCI_SW_START Configures whether or not to send data packets when the encoder state machine is in ST_SW_WAIT state.

0: Not send

1: Send

(R/W/SC)

Register 42.71. UHCI_ESCAPE_CONF_REG (0x0020)

[illegible]

UHCI_TX_CO_ESC_EN Configures whether or not to decode character 0xC0 when DMA receives data.

0: Not decode

1: Decode

(R/W)

UHCI_TX_DB_ESC_EN Configures whether or not to decode character 0xDB when DMA receives data.

0: Not decode

1: Decode

(R/W)

UHCI_TX_11_ESC_EN Configures whether or not to decode flow control character 0x11 when DMA receives data.

0: Not decode

1: Decode

(R/W)

UHCI_TX_13_ESC_EN	Configures whether or not to decode flow control character 0x13 when DMA receives data.
--------------------------	---

0: Not decode

1: Decode

(R/W)

UHCI_RX_CO_ESC_EN Configures whether or not to replace 0xC0 by special characters when DMA sends data.

0: Not replace

1: Replace

(R/W)

UHCI_RX_DB_ESC_EN Configures whether or not to replace 0xDB by special characters when DMA sends data.

0: Not replace

1: Replace

(R/W)

Continued on the next page...

Register 42.71. UHCI_ESCAPE_CONF_REG (0x0020)

Continued from the previous page...

UHCI_RX_11_ESC_EN Configures whether or not to replace flow control character 0x11 by special characters when DMA sends data.

0: Not replace

1: Replace

(R/W)

UHCI_RX_13_ESC_EN Configures whether or not to replace flow control character 0x13 by special characters when DMA sends data.

0: Not replace

1: Replace

(R/W)

Register 42.72. UHCI_HUNG_CONF_REG (0x0024)

(reserved)								UHCL_RXFIFO_TIMEOUT_ENA				UHCL_RXFIFO_TIMEOUT_SHIFT				UHCL_RXFIFO_TIMEOUT				UHCL_TXFIFO_TIMEOUT_ENA				UHCL_TXFIFO_TIMEOUT_SHIFT				UHCL_TXFIFO_TIMEOUT			
31							24	23	22		20	19				12	11	10		8	7							0			
0	0	0	0	0	0	0	0	1		0					0x10		1		0								0x10	Reset			

UHCI_TXFIFO_TIMEOUT Configures the timeout value for DMA data reception.

Measurement unit: ms. (R/W)

UHCI_TXFIFO_TIMEOUT_SHIFT Configures the upper limit of the timeout counter for TX FIFO. (R/W)

UHCI_TXFIFO_TIMEOUT_ENA Configures whether or not to enable the data reception timeout for TX FIFO.

0: Disable

1: Enable

(R/W)

UHCI_RXFIFO_TIMEOUT Configures the timeout value for DMA to read data from RAM.

Measurement unit: ms. (R/W)

UHCI_RXFIFO_TIMEOUT_SHIFT Configures the upper limit of the timeout counter for RX FIFO. (R/W)

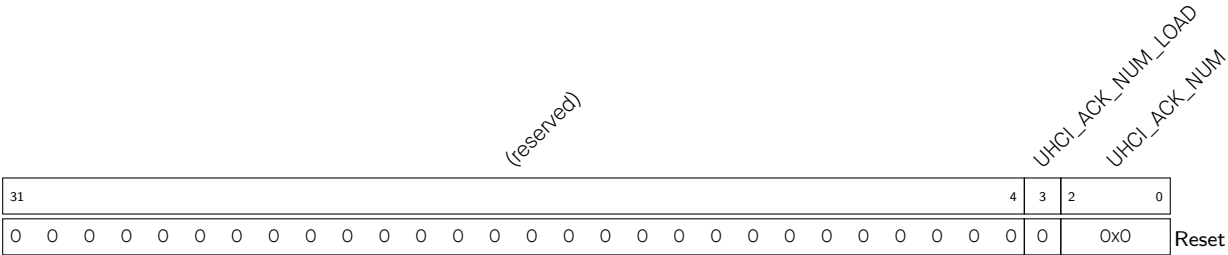
UHCI_RXFIFO_TIMEOUT_ENA Configures whether or not to enable the DMA data transmission timeout.

0: Disable

1: Enable

(R/W)

Register 42.73. UHCI_ACK_NUM_REG (0x0028)



UHCI_ACK_NUM Configures the number of acknowledgements used in software flow control.
(R/W)

UHCI_ACK_NUM_LOAD Configures whether or not load acknowledgements.
0: Not load
1: Load
(WT)

Register 42.74. UHCI_QUICK_SENT_REG (0x0030)

(reserved)																								UHCI_ALWAYS_SEND_EN				UHCI_ALWAYS_SEND_NUM				UHCI_SINGLE_SEND_EN				UHCI_SINGLE_SEND_NUM																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
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UHCI_SINGLE_SEND_NUM Configures the source of data to be transmitted in single_send mode.

- 0: Q0 register
 - 1: Q1 register
 - 2: Q2 register
 - 3: Q3 register
 - 4: Q4 register
 - 5: Q5 register
 - 6: Q6 register
 - 7: Invalid. No effect
- (R/W)

UHCI_SINGLE_SEND_EN Configures whether or not to enable single_send mode.

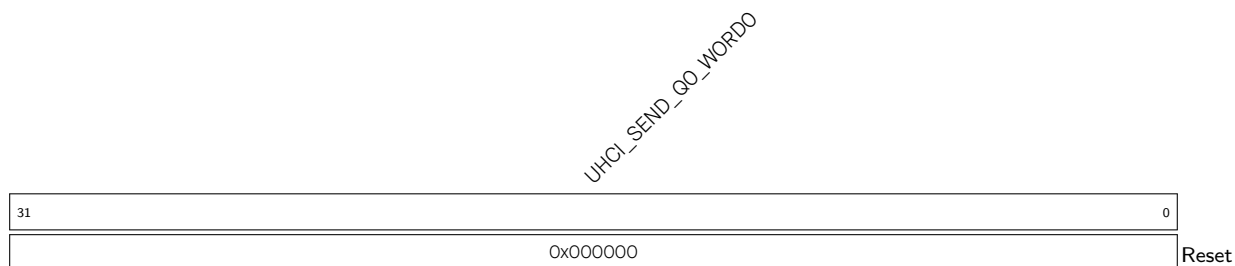
- 0: Disable
 - 1: Enable
- (R/W/SC)

UHCI_ALWAYS_SEND_NUM Configures the source of data to be transmitted in always_send mode.

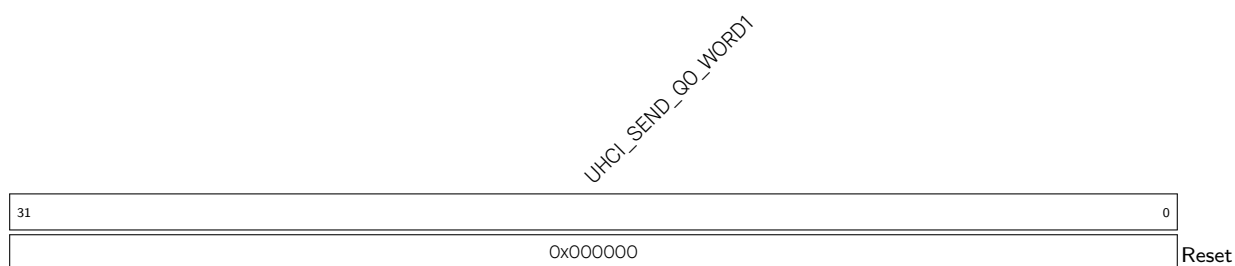
- 0: Q0 register
 - 1: Q1 register
 - 2: Q2 register
 - 3: Q3 register
 - 4: Q4 register
 - 5: Q5 register
 - 6: Q6 register
 - 7: Invalid. No effect
- (R/W)

UHCI_ALWAYS_SEND_EN Configures whether or not to enable always_send mode.

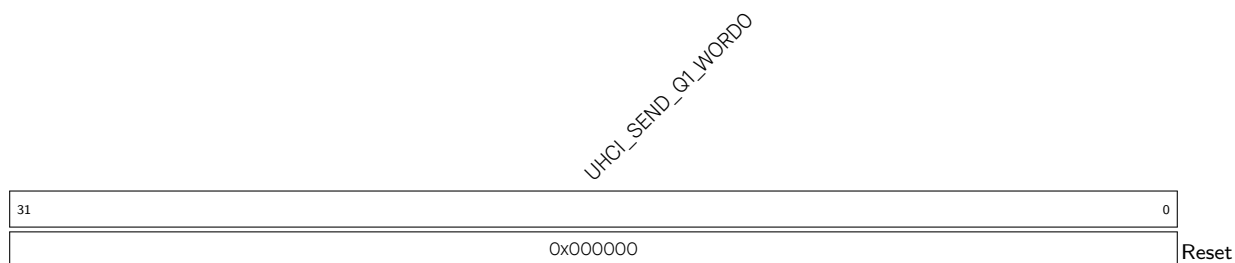
- 0: Disable
 - 1: Enable
- (R/W)

Register 42.75. UHCI_REG_Q0_WORD0_REG (0x0034)

UHCI_SEND_Q0_WORD0 Data to be transmitted in Q0 register. (R/W)

Register 42.76. UHCI_REG_Q0_WORD1_REG (0x0038)

UHCI_SEND_Q0_WORD1 Data to be transmitted in Q0 register. (R/W)

Register 42.77. UHCI_REG_Q1_WORD0_REG (0x003C)

UHCI_SEND_Q1_WORD0 Data to be transmitted in Q1 register. (R/W)

Register 42.78. UHCI_REG_Q1_WORD1_REG (0x0040)

<i>UHCI_SEND_Q1_WORD1</i>	
31	0
0x000000	
Reset	

UHCI_SEND_Q1_WORD1 Data to be transmitted in Q1 register. (R/W)

Register 42.79. UHCI_REG_Q2_WORD0_REG (0x0044)

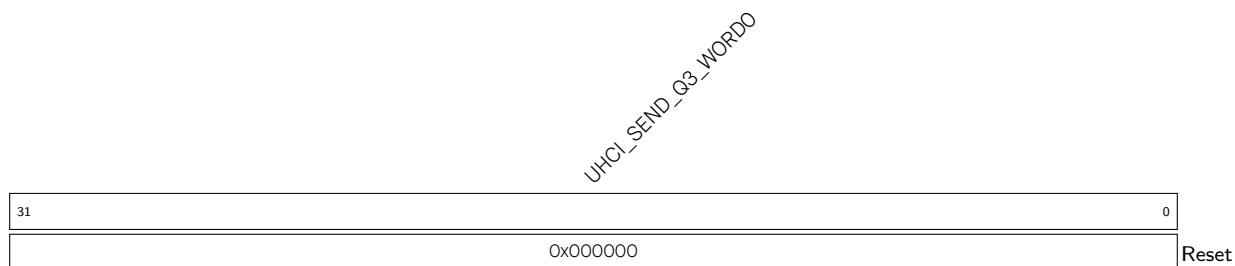
<i>UHCI_SEND_Q2_WORD0</i>	
31	0
0x000000	
Reset	

UHCI_SEND_Q2_WORD0 Data to be transmitted in Q2 register. (R/W)

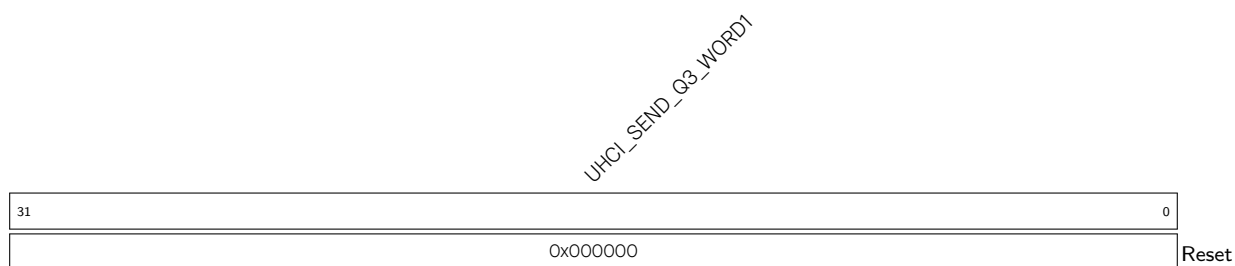
Register 42.80. UHCI_REG_Q2_WORD1_REG (0x0048)

<i>UHCI_SEND_Q2_WORD1</i>	
31	0
0x000000	
Reset	

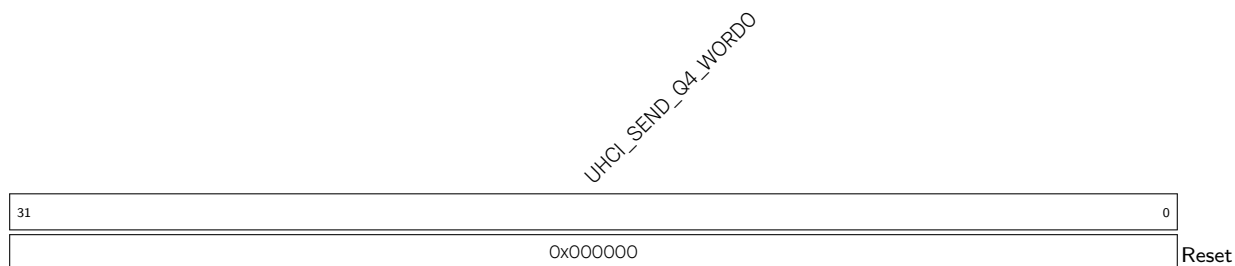
UHCI_SEND_Q2_WORD1 Data to be transmitted in Q2 register. (R/W)

Register 42.81. UHCI_REG_Q3_WORD0_REG (0x004C)

UHCI_SEND_Q3_WORD0 Data to be transmitted in Q3 register. (R/W)

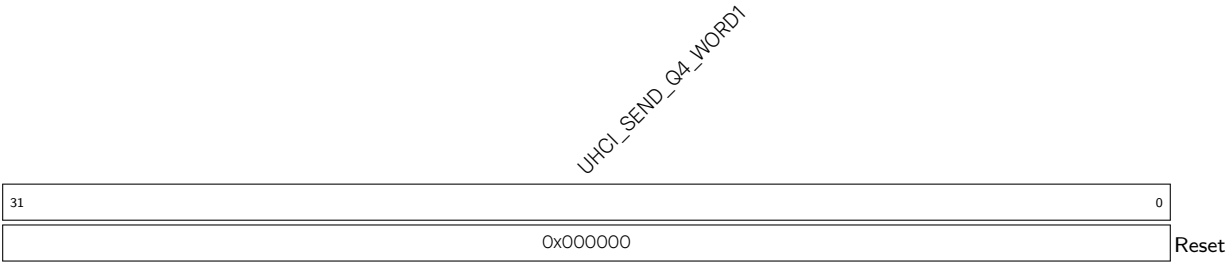
Register 42.82. UHCI_REG_Q3_WORD1_REG (0x0050)

UHCI_SEND_Q3_WORD1 Data to be transmitted in Q3 register. (R/W)

Register 42.83. UHCI_REG_Q4_WORD0_REG (0x0054)

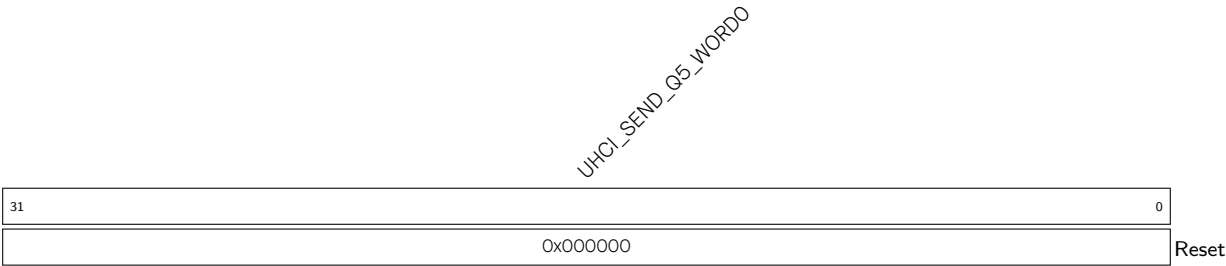
UHCI_SEND_Q4_WORD0 Data to be transmitted in Q4 register. (R/W)

Register 42.84. UHCI_REG_Q4_WORD1_REG (0x0058)



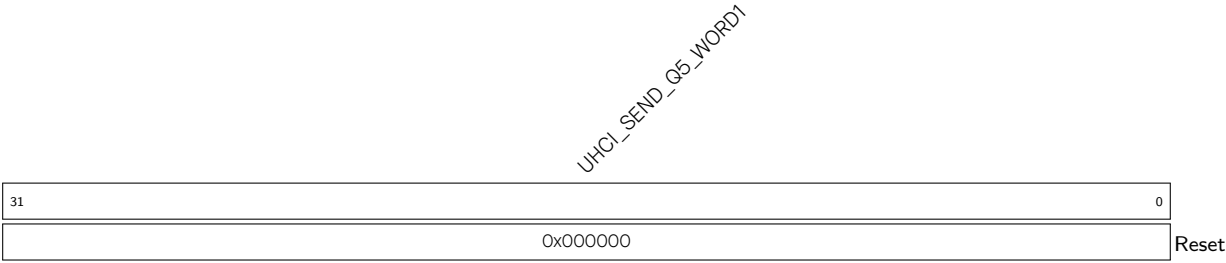
UHCI_SEND_Q4_WORD1 Data to be transmitted in Q4 register. (R/W)

Register 42.85. UHCI_REG_Q5_WORD0_REG (0x005C)



UHCI_SEND_Q5_WORD0 Data to be transmitted in Q5 register. (R/W)

Register 42.86. UHCI_REG_Q5_WORD1_REG (0x0060)



UHCI_SEND_Q5_WORD1 Data to be transmitted in Q5 register. (R/W)

Register 42.87. UHCI_REG_Q6_WORD0_REG (0x0064)

<i>UHCI_SEND_Q6_WORD0</i>	
31	0
0x000000	
Reset	

UHCI_SEND_Q6_WORD0 Data to be transmitted in Q6 register. (R/W)

Register 42.88. UHCI_REG_Q6_WORD1_REG (0x0068)

<i>UHCI_SEND_Q6_WORD1</i>	
31	0
0x000000	
Reset	

UHCI_SEND_Q6_WORD1 Data to be transmitted in Q6 register. (R/W)

Register 42.89. UHCI_ESC_CONFO_REG (0x006C)

(reserved)								UHCI_SEPER_ESC_CHAR1								UHCI_SEPER_ESC_CHAR0								UHCI_SEPER_CHAR								
3124								2316								158								70								
00000000								Oxdc								Oxdb								Oxc0								Reset

UHCI_SEPER_CHAR Configures separators to encode data packets. The default value is 0xc0. (R/W)

UHCI_SEPER_ESC_CHAR0 Configures the first character of SLIP escape sequence. The default value is 0xdb. (R/W)

UHCI_SEPER_ESC_CHAR1 Configures the second character of SLIP escape sequence. The default value is 0xdc. (R/W)

Register 42.90. UHCI_ESC_CONF1_REG (0x0070)

(reserved)								UHCI_ESC_SEQ0_CHAR1								UHCI_ESC_SEQ0_CHAR0								UHCI_ESC_SEQ0																								
31								24								16								15								8								7								0
0	0	0	0	0	0	0	0		0xdd								0xdb								0xdb								Reset															

UHCI_ESC_SEQ0 Configures the character that needs to be encoded. The default value is 0xDB used as the first character of SLIP escape sequence. (R/W)

UHCI_ESC_SEQ0_CHAR0 Configures the first character of SLIP escape sequence. The default value is 0xDB. (R/W)

UHCI_ESC_SEQ0_CHAR1 Configures the second character of SLIP escape sequence. The default value is 0xDD. (R/W)

Register 42.91. UHCI_ESC_CONF2_REG (0x0074)

(reserved)								UHCI_ESC_SEQ1_CHAR1								UHCI_ESC_SEQ1_CHAR0								UHCI_ESC_SEQ1								
3124								2316								158								70								
00000000								0xde								0xdb								0x11								Reset

UHCI_ESC_SEQ1 Configures a character that need to be encoded. The default value is 0x11 used as a flow control character. (R/W)

UHCI_ESC_SEQ1_CHAR0 Configures the first character of SLIP escape sequence. The default value is 0xDB. (R/W)

UHCI_ESC_SEQ1_CHAR1 Configures the second character of SLIP escape sequence. The default value is 0xDE. (R/W)

Register 42.92. UHCI_ESC_CONF3_REG (0x0078)

(reserved)								UHCL_ESC_SEQ2_CHAR1								UHCL_ESC_SEQ2_CHAR0								UHCL_ESC_SEQ2																							
31								24								16								8								7								0							
0 0 0 0 0 0 0 0								Oxdf								Oxdb								0x13								Reset															

UHCI_ESC_SEQ2 Configures the character that needs to be decoded. The default value is 0x13 used as a flow control character. (R/W)

UHCI_ESC_SEQ2_CHAR0 Configures the first character of SLIP escape sequence. The default value is 0xDB. (R/W)

UHCI_ESC_SEQ2_CHAR1 Configures the second character of SLIP escape sequence. The default value is 0xDF. (R/W)

Register 42.93. UHCI_PKT_THRES_REG (0x007C)

(reserved)																UHOI_PKT_THRS															
31																0															
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0x80															
																Reset															

UHCL_PKT_THRS Configures the maximum value of the packet length.
Measurement unit: byte.
Valid only when UHCL_HEAD_EN is 0. (R/W)

Register 42.94. UHCI_INT_RAW_REG (0x0004)

(reserved)																								UHCI_APP_CTRL1_INT_RAW UHCI_APP_CTRL0_INT_RAW UHCI_OUT_EOF_INT_RAW UHCI_SEND_A_REG_Q_INT_RAW UHCI_SEND_S_REG_Q_INT_RAW UHCI_TX_HUNG_INT_RAW UHCI_RX_HUNG_INT_RAW UHCI_TX_START_INT_RAW UHCI_RX_START_INT_RAW																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
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UHCI_RX_START_INT_RAW The raw interrupt status of UHCI_RX_START_INT. (R/WTC/SS)

UHCI_TX_START_INT_RAW The raw interrupt status of UHCI_TX_START_INT. (R/WTC/SS)

UHCI_RX_HUNG_INT_RAW The raw interrupt status of UHCI_RX_HUNG_INT. (R/WTC/SS)

UHCI_TX_HUNG_INT_RAW The raw interrupt status of UHCI_TX_HUNG_INT. (R/WTC/SS)

UHCI_SEND_S_REG_Q_INT_RAW The raw interrupt status of UHCI_SEND_S_REG_Q_INT.
(R/WTC/SS)

UHCI_SEND_A_REG_Q_INT_RAW The raw interrupt status of UHCI_SEND_A_REG_Q_INT.
(R/WTC/SS)

UHCI_OUT_EOF_INT_RAW The raw interrupt status of UHCI_OUT_EOF_INT. (R/WTC/SS)

UHCI_APP_CTRL0_INT_RAW The raw interrupt status of UHCI_APP_CTRL0_INT. (R/W)

UHCI_APP_CTRL1_INT_RAW The raw interrupt status of UHCI_APP_CTRL1_INT. (R/W)

Register 42.95. UHCI_INT_ST_REG (0x0008)

[illegible]

UHCI_RX_START_INT_ST The masked interrupt status of UHCI_RX_START_INT. (RO)

UHCI_TX_START_INT_ST	The masked interrupt status of UHCI_TX_START_INT. (RO)
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UHCI_RX_HUNG_INT_ST	The masked interrupt status of UHCI_RX_HUNG_INT. (RO)
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UHCI_TX_HUNG_INT_ST The masked interrupt status of UHCI_TX_HUNG_INT. (RO)

UHCI_SEND_S_REG_Q_INT_ST The masked interrupt status of UHCI_SEND_S_REG_Q_INT. (RO)

UHCI_SEND_A_REG_Q_INT_ST The masked interrupt status of UHCI_SEND_A_REG_Q_INT. (RO)

UHCI_OUTLINK_EOF_ERR_INT_ST The masked interrupt status of UHCI_OUTLINK_EOF_ERR_INT.
(RO)

UHCI_APP_CTRL0_INT_ST The masked interrupt status of UHCI_APP_CTRL0_INT. (RO)

UHCI_APP_CTRL1_INT_ST The masked interrupt status of UHCI_APP_CTRL1_INT. (RO)

Register 42.96. UHCI_INT_ENA_REG (0x000C)

(reserved)																								UHCI_APP_CTRL1_INT_ENA UHCI_APP_CTRL0_INT_ENA UHCI_OUTLINK_EOF_ERR_INT_ENA UHCI_SEND_A_REG_Q_INT_ENA UHCI_SEND_S_REG_Q_INT_ENA UHCI_TX_HUNG_INT_ENA UHCI_RX_HUNG_INT_ENA UHCI_TX_START_INT_ENA UHCI_RX_START_INT_ENA								
31																	9	8	7	6	5	4	3	2	1	0	Reset					
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													

UHCI_RX_START_INT_ENA Write 1 to enable UHCI_RX_START_INT. (R/W)

UHCI_TX_START_INT_ENA Write 1 to enable UHCI_TX_START_INT. (R/W)

UHCI_RX_HUNG_INT_ENA Write 1 to enable UHCI_RX_HUNG_INT. (R/W)

UHCI_TX_HUNG_INT_ENA Write 1 to enable UHCI_TX_HUNG_INT. (R/W)

UHCI_SEND_S_REG_Q_INT_ENA Write 1 to enable UHCI_SEND_S_REG_Q_INT. (R/W)

UHCI_SEND_A_REG_Q_INT_ENA Write 1 to enable UHCI_SEND_A_REG_Q_INT. (R/W)

UHCI_OUTLINK_EOF_ERR_INT_ENA Write 1 to enable UHCI_OUTLINK_EOF_ERR_INT. (R/W)

UHCI_APP_CTRL0_INT_ENA Write 1 to enable UHCI_APP_CTRL0_INT. (R/W)

UHCI_APP_CTRL1_INT_ENA Write 1 to enable UHCI_APP_CTRL1_INT. (R/W)

Register 42.97. UHCI_INT_CLR_REG (0x0010)

(reserved)																UHCI_APP_CTRL1_INT_CLR			
																UHCI_APP_CTRL0_INT_CLR			
																UHCI_OUTLINK_EOF_ERR_INT_CLR			
																UHCI_SEND_A_REG_Q_INT_CLR			
																UHCI_SEND_S_REG_Q_INT_CLR			
																UHCI_TX_HUNG_INT_CLR			
																UHCI_RX_HUNG_INT_CLR			
																UHCI_TX_START_INT_CLR			
																UHCI_RX_START_INT_CLR			
31									9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

UHCI_RX_START_INT_CLR Write 1 to clear UHCI_RX_START_INT. (WT)

UHCI_TX_START_INT_CLR Write 1 to clear UHCI_TX_START_INT. (WT)

UHCI_RX_HUNG_INT_CLR Write 1 to clear UHCI_RX_HUNG_INT. (WT)

UHCI_TX_HUNG_INT_CLR Write 1 to clear UHCI_TX_HUNG_INT. (WT)

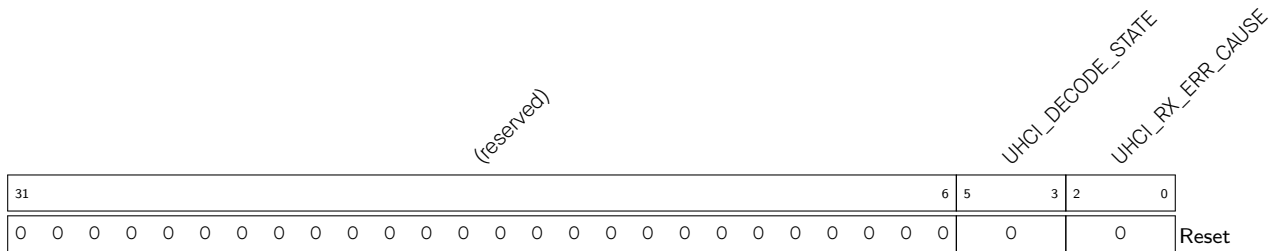
UHCI_SEND_S_REG_Q_INT_CLR Write 1 to clear UHCI_SEND_S_REG_Q_INT. (WT)

UHCI_SEND_A_REG_Q_INT_CLR Write 1 to clear UHCI_SEND_A_REG_Q_INT. (WT)

UHCI_OUTLINK_EOF_ERR_INT_CLR Write 1 to clear UHCI_OUTLINK_EOF_ERR_INT. (WT)

UHCI_APP_CTRL0_INT_CLR Write 1 to clear UHCI_APP_CTRL0_INT. (WT)

UHCI_APP_CTRL1_INT_CLR Write 1 to clear UHCI_APP_CTRL1_INT. (WT)

Register 42.98. UHCI_STATE0_REG (0x0018)

UHCI_RX_ERR_CAUSE Represents the error type when DMA has received a packet with error.

0: Invalid. No effect

1: Checksum error in the HCI packet

2: Sequence number error in the HCI packet

3: CRC bit error in the HCI packet

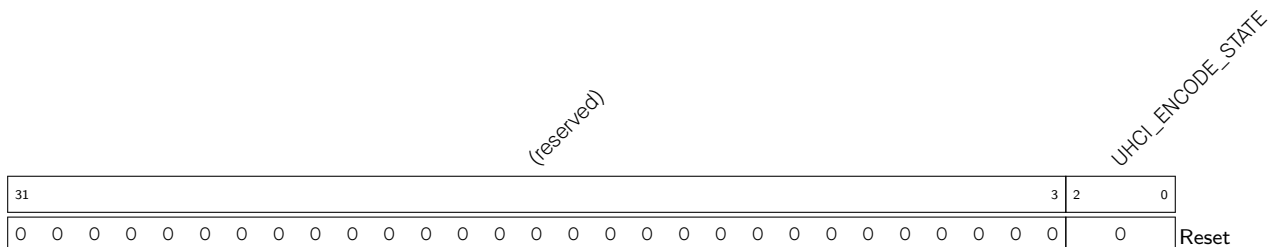
4: 0xC0 is found but the received HCI packet is not complete 5: 0xC0 is not found when the HCI packet has been received

6: CRC check error

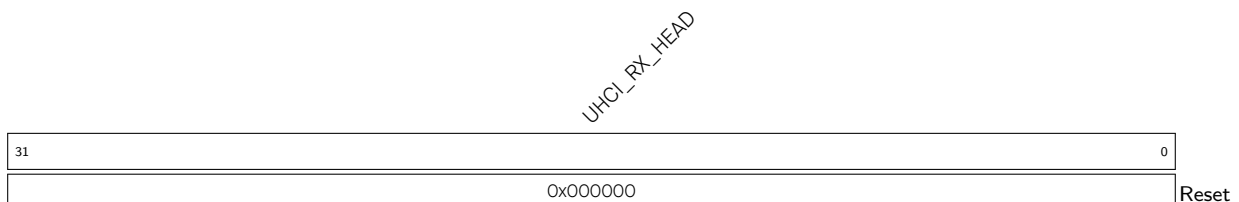
7: Invalid. No effect

(RO)

UHCI_DECODE_STATE Represents the UHCI decoder status. (RO)

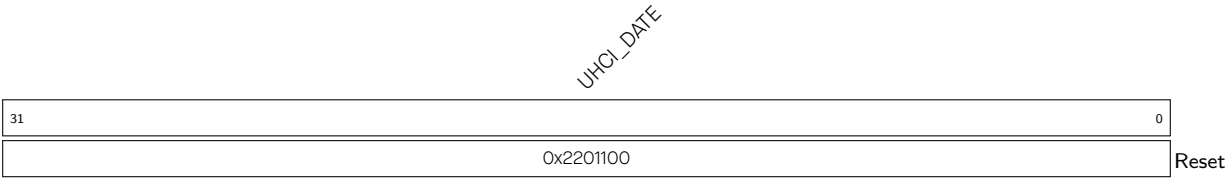
Register 42.99. UHCI_STATE1_REG (0x001C)

UHCI_ENCODE_STATE Represents the UHCI encoder status. (RO)

Register 42.100. UHCI_RX_HEAD_REG (0x002C)

UHCI_RX_HEAD Represents the header of the current received packet. (RO)

Register 42.101. UHCI_DATE_REG (0x0080)



UHCI_DATE Version control register. (R/W)